

★ WEEK 6 · PART 1 of 3

The VAR Model

When Variables Move Together

Vector autoregression · the coefficient matrix · stability · joint forecasting

Univariate models treat each series in isolation, but markets are interconnected. Vector autoregression lets several series interact — each one's future shaped by every series' past.

1

PART 1 · THE VAR MODEL

ARMA, for a Vector of Series

Why we model series jointly, the VAR(p) equation and its coefficient matrix, the stability condition, lag selection, and forecasting several variables at once.

The Man Who Told Economists to Stop Pretending — Part 1

A 1980 paper that overturned macro modeling

Christopher Sims argued that the huge structural macro models of the day rested on 'incredible' assumptions — theory dressed up as fact. He proposed letting the data speak instead.

The problem he saw

- Big models imposed hundreds of untested theoretical restrictions.
- Those restrictions were assumed, not evidenced.
- Forecasts and policy advice inherited every hidden assumption.

His alternative

- Treat all variables symmetrically — none privileged by theory.
- Let each depend on the recent past of them all.
- That minimal-assumption model is the VAR.

The Man Who Told Economists to Stop Pretending — Part 2

The VAR revolution, and the lesson

Sims' VAR became the workhorse of empirical macro and finance, and earned him the 2011 Nobel. Its lesson: model the relationships you can measure, not the ones you assume.

What it changed

- VAR became the standard tool for studying co-movement.
- Impulse responses and causality tests grew out of it.
- It reshaped how economists read data — humbly.

The lesson for us

- Let several series interact, with minimal theory imposed.
- Measure the dynamics; don't assume them.
- But stay humble about what 'causality' really means (Part 2).

Why Go Multivariate

Three reasons one-at-a-time is not enough

1

Markets are interconnected

Rates move stocks; oil moves airlines; a shock to one asset ripples into others. A model of one series alone is blind to all of it.

2

Univariate models miss the links

Even our GARCH work (Week 5) modeled each series' risk in isolation. Portfolios and pairs depend on how series move together.

3

VAR captures the feedback

By letting every variable depend on every variable's past, a VAR models lead-lag, feedback, and co-movement in one system.

Part 1 in one line: stop modeling one series alone — let several interact in a single system.

Why Model Series Together?

Isolated models miss the connections that matter

When variables influence each other, modeling them separately throws away information. A joint model uses one series' past to help forecast another — and to study how shocks spread.

What you gain

- Better forecasts, by borrowing strength across series.
- A way to test who leads whom (Granger causality, Part 2).
- A map of how a shock propagates through the system (IRF, Part 3).

Everyday examples

- Short and long interest rates that move together.
- A stock and its sector index.
- Spot and futures prices — the seed of a pairs trade.

The VAR(p) Model

ARMA, but for a vector of variables

$$\mathbf{Y}_t = \mathbf{c} + A_1 \mathbf{Y}_{t-1} + A_2 \mathbf{Y}_{t-2} + \cdots + A_p \mathbf{Y}_{t-p} + \boldsymbol{\varepsilon}_t \quad \text{VAR}(p)$$

Reading it

- $\mathbf{Y}_t$ is a VECTOR — several series stacked together.
- Each A is a MATRIX of coefficients, one per lag.
- Every variable regresses on the past of ALL variables.

Why it generalizes AR

- With one variable it collapses to an ordinary AR(p).
- The matrices A hold both own-effects and cross-effects.
- p is the lag order, exactly as before.

A Bivariate VAR(1), Written Out

Two equations make the idea plain

$$\begin{aligned}y_{1,t} &= c_1 + a_{11}y_{1,t-1} + a_{12}y_{2,t-1} + \varepsilon_{1,t} \\ y_{2,t} &= c_2 + a_{21}y_{1,t-1} + a_{22}y_{2,t-1} + \varepsilon_{2,t}\end{aligned}$$

The two equations

- Each series depends on its OWN last value (a_{11} , a_{22}).
- And on the OTHER series' last value (a_{12} , a_{21}).
- Two coupled equations, estimated together.

Where the magic is

- $a_{12} \neq 0$ means Y2's past helps predict Y1.
- $a_{21} \neq 0$ means Y1's past helps predict Y2.
- Those cross-terms are the whole point of a VAR.

Reading the Coefficient Matrix

Where the cross-dynamics live

$$A_1 = \begin{pmatrix} a_{11} & \overbrace{a_{12}}^{\text{CROSS}} \\ \underbrace{a_{21}}_{\text{CROSS}} & a_{22} \end{pmatrix} \quad \text{off-diagonals carry the interactions}$$

Diagonal vs off-diagonal

- Diagonal terms are the usual own-lag (AR) effects.
- Off-diagonal terms carry the interactions between series.
- A diagonal matrix means NO interaction — separate ARs.

How to read it

- A large off-diagonal entry = a strong lead-lag link.
- Its sign tells you the direction of the influence.
- The whole matrix is a compact map of the system.

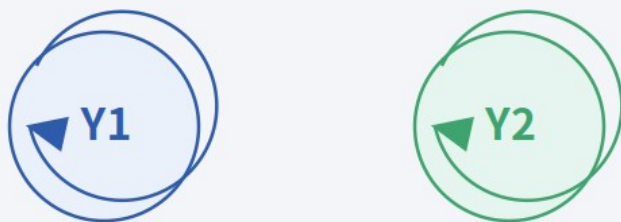
Univariate vs VAR: Who Listens to Whom

Separate models let each series hear only itself; a VAR lets every series hear every series' past — so feedback and co-movement are finally captured.

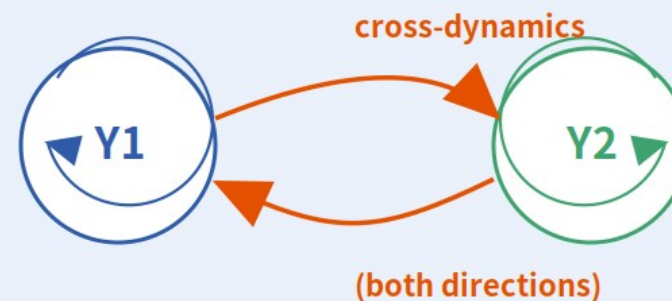
Univariate vs VAR: Who Listens to Whom

In a VAR, every variable's forecast uses every variable's past - the links univariate models ignore.

Two univariate models
each series listens only to ITSELF



One VAR
each series listens to BOTH series' past



feedback and co-movement are captured

What a VAR Captures

Feedback, lead-lag, and co-movement

The dynamics it sees

- Feedback loops: X moves Y, which later moves X back.
- Lead-lag: one series systematically anticipates another.
- Common responses to shared, underlying forces.

Why it matters in finance

- Risk and return spill across related assets.
- Hedges and pairs depend on these joint dynamics.
- A single-series model literally cannot represent them.

The Reduced Form: Let the Data Speak

Minimal theory, maximal honesty

A reduced-form VAR imposes almost no economic theory — just that variables depend on their recent joint past. It measures relationships rather than assuming them.

What 'reduced form' means

- No variable is singled out as the cause of another.
- The equations describe correlations through time, not mechanisms.
- It is deliberately agnostic about the 'true' structure.

The trade-off

- You get honest, data-driven dynamics and forecasts.
- But you cannot read causal 'shocks' off it directly.
- Recovering those needs extra assumptions (Part 3).

Estimating a VAR

Surprisingly simple: one equation at a time

The method

- Each equation has the SAME regressors — all lagged variables.
- So you can estimate each one by ordinary least squares.
- Equation-by-equation OLS is efficient here.

What you get

- The coefficient matrices A and the residual covariance.
- That residual covariance matters later (correlated shocks, Part 3).
- Software fits the whole system in one call.

Stability of a VAR

The multivariate version of the unit-root condition

stable $\iff$ all eigenvalues of the companion matrix lie INSIDE the unit circle

The condition

- Stack the VAR into a big 'companion' matrix.
- All its eigenvalues must lie inside the unit circle.
- For one variable this reduces to $|\phi| < 1$.

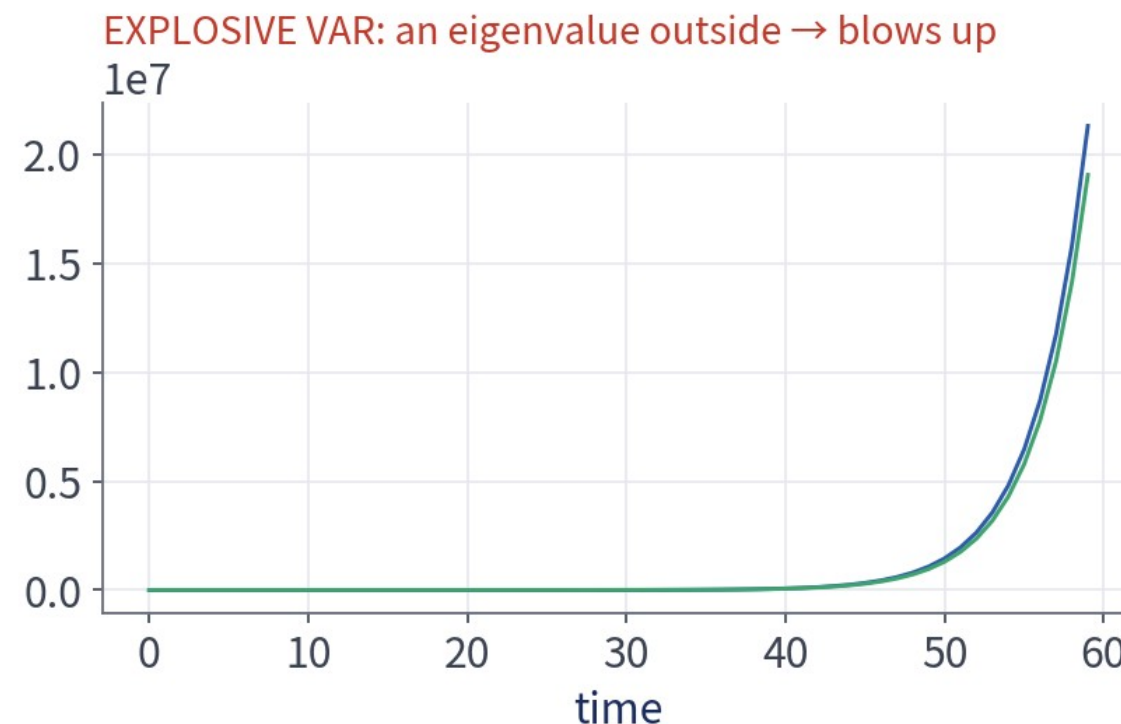
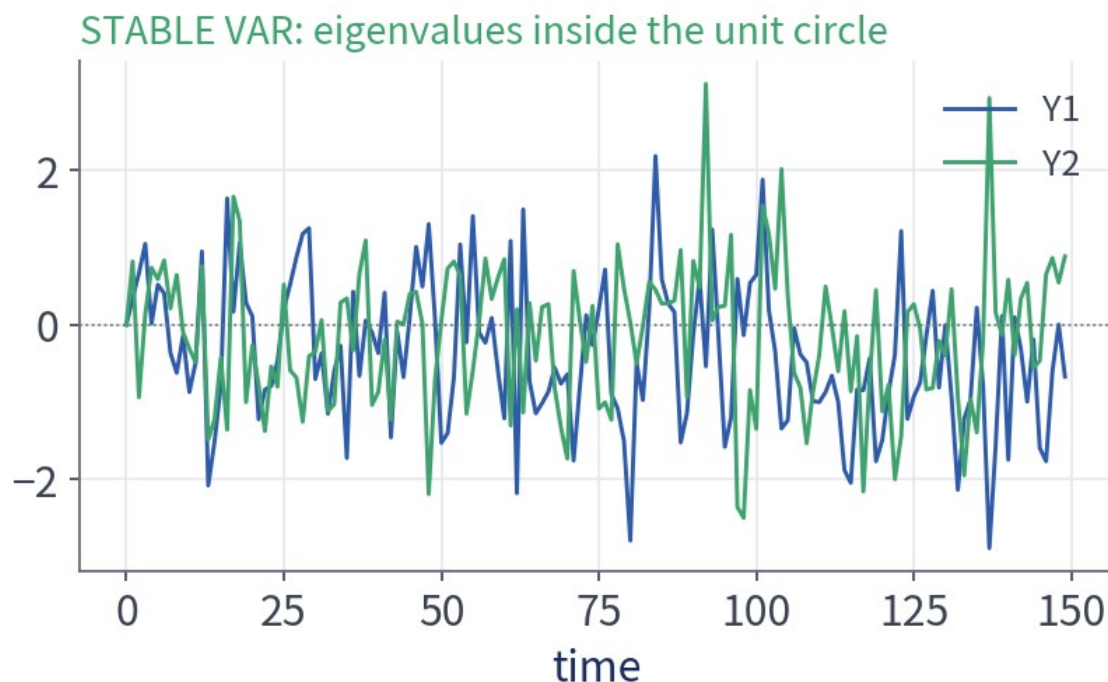
Why it matters

- A stable VAR is stationary — its shocks eventually fade.
- An unstable VAR explodes, exactly like a unit root.
- Check stability before trusting any forecast or IRF.

Stable vs Explosive

With all eigenvalues inside the unit circle (left), the system fluctuates around a fixed level. Push one outside (right) and the whole system blows up.

Stability is the multivariate unit-root condition: all eigenvalues must lie inside the circle



Stationarity Still Comes First

Apply a VAR to stationary series

Like ARMA, a standard VAR assumes stationary inputs. If the series trend, difference them first — or, if they move together, a special structure awaits (Week 7).

The usual prep

- Test each series for a unit root (Week 2).
- Difference trending series to returns or growth rates.
- Then fit the VAR on the stationary transforms.

A tantalizing exception

- If two $I(1)$ series are cointegrated, differencing loses information.
- Their long-run link deserves its own model.
- That is the vector error-correction model — Week 7.

Choosing the Lag Order p

Balancing fit against an explosion of parameters

choose $p = \arg \min \text{ AIC / BIC / HQ; } \quad \# \text{params} = k + p k^2$

How to choose

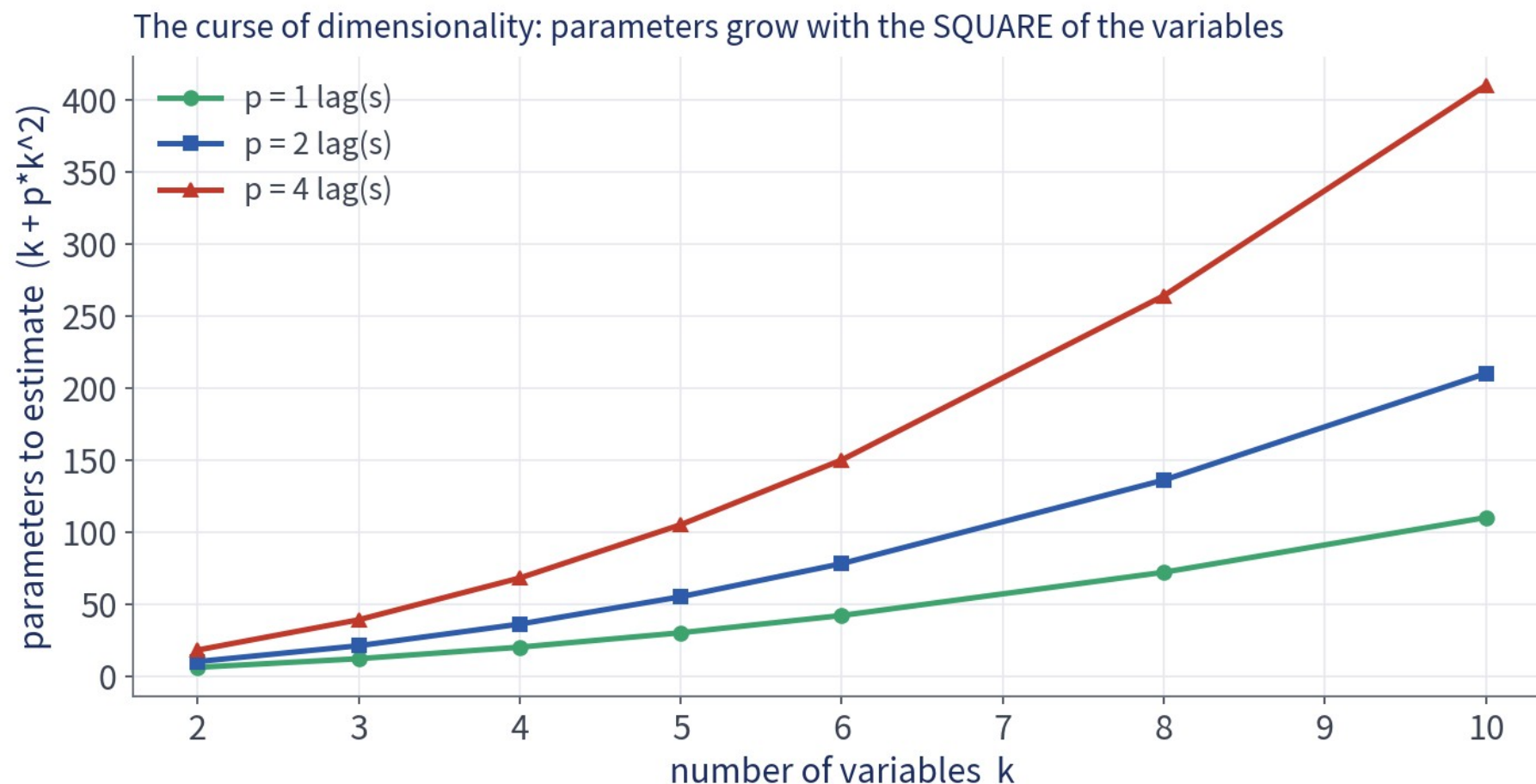
- Fit several lag orders and compare AIC, BIC, or HQ.
- Lower is better; BIC leans toward fewer lags.
- Confirm the residuals show no leftover autocorrelation.

The dimensionality trap

- Parameters grow with the SQUARE of the variable count.
- A few variables and a few lags is already a lot to estimate.
- Parsimony matters even more than in the univariate case.

The Curse of Dimensionality

The parameter count grows like $k^2 \cdot p$. Doubling the number of variables roughly quadruples the coefficients — which is why VARs are kept small and low-order.



Forecasting With a VAR

Many variables, forecast jointly

$$\hat{\mathbf{Y}}_{t+1} = \mathbf{c} + A_1 \mathbf{Y}_t + \cdots + A_p \mathbf{Y}_{t-p+1} \quad (\text{iterate for } h > 1)$$

How it works

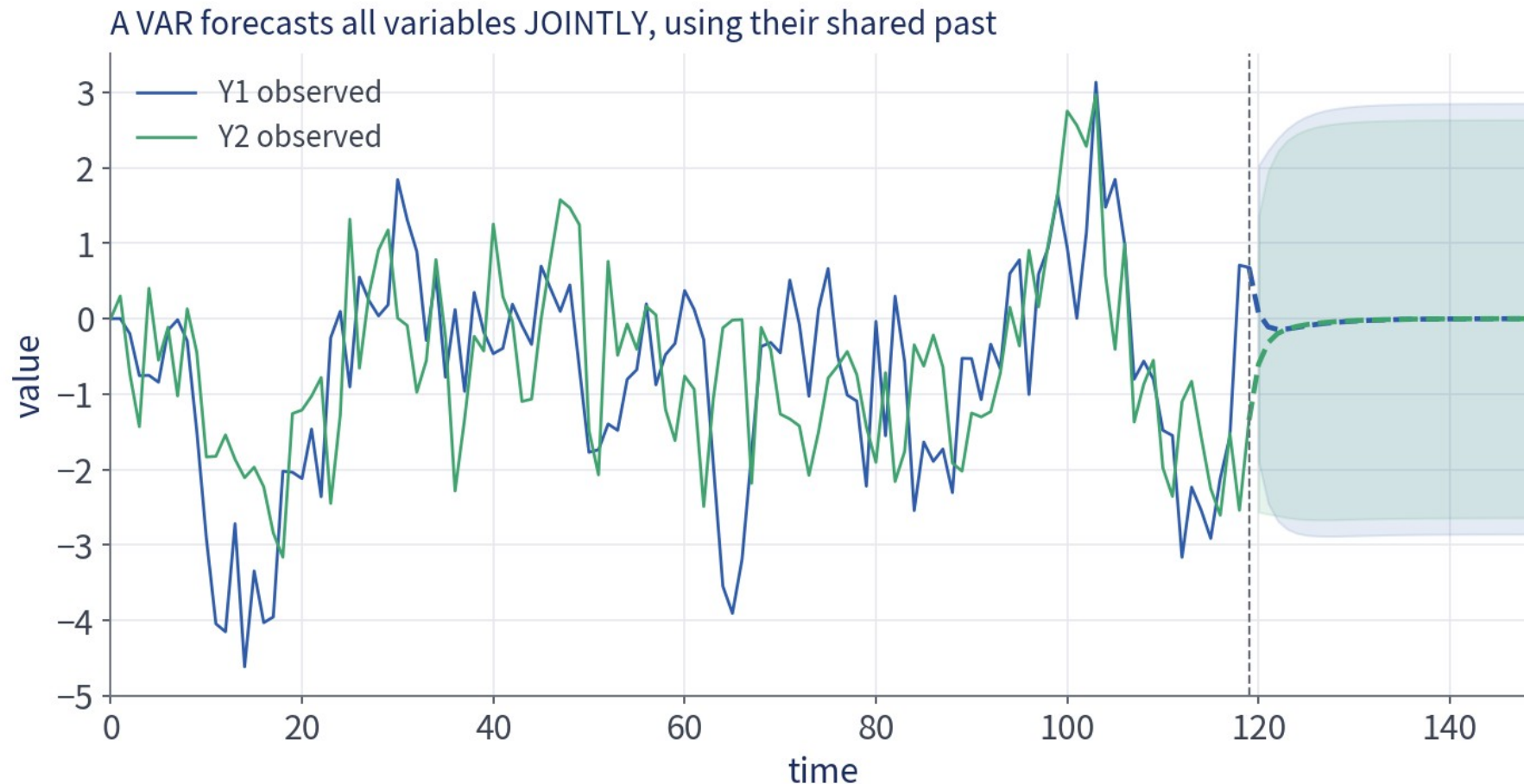
- One step ahead uses the last observed values of all series.
- Longer horizons feed the forecasts back in, iterating.
- Every series is forecast at once, consistently.

Why joint helps

- Each forecast exploits the others' recent moves.
- The forecasts respect the cross-dynamics you estimated.
- Intervals widen with horizon, as always — but jointly.

A VAR Forecasts Everything at Once

Both series are projected jointly, each using the other's recent past. The intervals fan out with the horizon, reflecting the accumulating shocks.



The VAR Workflow

Five steps from several series to a fitted system

1 · Stationarize

each series

Test and difference so every series is stationary.

2 · Select p

lag order

Compare AIC/BIC/HQ; keep p small.

3 · Estimate

OLS system

Fit equation-by-equation; get A and the residuals.

4 · Check

stability

Confirm eigenvalues inside the circle; clean residuals.

5 · Use

forecast+analyze

Forecast jointly; then Granger, IRF, FEVD.

Steps 1–4 build the model; step 5 — the interesting part — is Parts 2 and 3.

VAR vs Separate Univariate Models

What the joint model buys, and costs

Aspect	Separate models	VAR (joint)
Cross-dynamics	ignored	captured via off-diagonal terms
Forecasts	each series alone	all series jointly and consistently
Lead-lag / feedback	invisible	explicit and testable
Parameter cost	few	many — grows with k squared

VAR buys realism about interactions at the price of far more parameters — so keep k and p small.

Fit a VAR and Forecast

Direct Claude Code from several series to a joint model

GOAL Build a VAR on two or three related series, choose the lag order, check stability, and forecast them jointly.

[INPUT] Prompt · Claude Code

Write and run `var_lab.py` that:

1. downloads 2-3 related series (e.g. a stock, its sector ETF, and a rate) and makes each stationary (returns / differences);
 2. selects the VAR lag order p by AIC and BIC with `statsmodels`, and fits the VAR;
 3. checks stability (eigenvalues) and residual autocorrelation;
 4. forecasts all series 20 steps ahead with joint confidence bands, and plots them.
- Report the chosen p and whether it is stable.

How to work with it

- Confirm every series is stationary first.
- Watch how AIC and BIC can disagree on p .
- All eigenvalues inside the circle = stable.
- Keep it; Part 2 adds Granger causality.

Deliverable: `var_lab.py` + a joint forecast of your chosen series.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1

In a bivariate VAR, what does a non-zero off-diagonal coefficient mean?

One series predicts the other

Q2

How do you check whether a fitted VAR is stable?

Eigenvalues inside circle

Q3

Why do VARs force you to keep the number of variables small?

Params grow with k^2

All three come from the coefficient matrix, stability, and the dimensionality cost.

Three Misconceptions to Avoid

Where VAR intuition slips

Common myths

- "More variables always make a VAR better."
- "A VAR reveals what causes what."
- "You can fit a VAR straight to price levels."

The reality

- Each added variable multiplies the parameters — VARs stay small.
- A reduced-form VAR shows prediction, not causation (Part 2).
- Standard VARs need stationary inputs — difference first (or Week 7).

Three Things to Carry Forward

Part 1 in one screen

1

A VAR is ARMA for a vector

Every series depends on the recent past of all series. The off-diagonal coefficients carry the cross-dynamics that univariate models throw away.

2

Stability and parsimony are the constraints

All eigenvalues must sit inside the unit circle, and the parameter count grows with k^2 — so keep the system small and stationary.

3

The joint model unlocks new questions

With a fitted VAR we can forecast jointly and then ask who leads whom and how shocks propagate — the subjects of Parts 2 and 3.

Next — Part 2: Granger causality, testing whether one series' past truly helps predict another.

★ WEEK 6 · PART 1 · PRACTICE

The VAR Model, Worked by Hand

Forecast Two Series Jointly, Then Check It Won't Explode

A bivariate VAR(1), its stability, and its lag order

Part 1's core model, with the arithmetic shown. We write out a bivariate VAR(1) and forecast both series together - each borrowing the other's history - then check stability through the eigenvalues and pick the lag order with AIC and BIC.

1P

PART 1 · APPLICATIONS

Two Series, One Joint Model

Case A: a bivariate VAR(1) - write the two equations, then forecast both series one and two steps ahead, watching the cross-dynamics at work. Case B: check stability through the eigenvalues and choose the lag order with AIC and BIC. Then a Claude Code lab.

What Part 1 Told Us

Now we apply it - with numbers

1

A VAR models series together

Each variable depends on the recent past of every variable - so feedback, lead-lag, and co-movement are all captured at once.

2

The coefficient matrix holds the cross-dynamics

Its off-diagonal entries are how one series influences another. Forecasting is one matrix-vector multiply per step.

3

Stability and lag order come first

The eigenvalues must sit inside the unit circle, and the lag order p balances fit against a blow-up in parameters.

Everything here is a calculator exercise: one forecast, one eigenvalue check, one lag choice.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from the coefficient matrix.
- Forecast, or check stability and lag.
- Then read what the numbers say.

What you should watch

- How the cross term feeds one series into another.
- How the eigenvalues decide stability.
- How AIC and BIC can disagree.

A Bivariate VAR(1)

Forecast two returns together

A stock's monthly return and its sector's return move together. A VAR(1) captures that by letting each depend on both series' last values. Given the coefficient matrix and last month's returns, we forecast both one and two steps ahead - and see the cross term carry the sector's move into the stock's forecast

The setup

- Two returns: a stock and its sector.
- Last month: stock 2.0%, sector 1.0%.
- A given coefficient matrix and constant.

What we'll do by hand

- Write the two equations out.
- Forecast one step with a matrix multiply.
- Iterate for the two-step forecast.

The VAR(1), Written Out

Each equation borrows the other's past

$$\begin{aligned} y_{1,t} &= 0.2 + 0.5 y_{1,t-1} + \underbrace{0.3 y_{2,t-1}}_{\text{cross term}} + \varepsilon_{1,t} \\ y_{2,t} &= 0.1 + 0.1 y_{1,t-1} + 0.4 y_{2,t-1} + \varepsilon_{2,t} \end{aligned}$$

Each series depends on both series' last values. The cross term - 0.3 times the sector in the stock's equation - is where the multivariate payoff lives: the stock's forecast uses the sector's history.

The One-Step Forecast

One matrix-vector multiply

$$\hat{y}_t = c + A y_{t-1} = \begin{bmatrix} 0.2 \\ 0.1 \end{bmatrix} + \begin{bmatrix} 0.5 & 0.3 \\ 0.1 & 0.4 \end{bmatrix} \begin{bmatrix} 2.0 \\ 1.0 \end{bmatrix} = \begin{bmatrix} 1.50 \\ 0.70 \end{bmatrix}$$

The stock

- $0.2 + 0.5(2.0) + 0.3(1.0)$.
- Its own momentum plus the sector's pull.
- Forecast: 1.50%.

The sector

- $0.1 + 0.1(2.0) + 0.4(1.0)$.
- A little from the stock, more from itself.
- Forecast: 0.70%.

The Two-Step Forecast

Feed the forecast back in

$$\hat{y}_{t+1} = c + A \hat{y}_t : \quad \hat{y}_1 = 0.2 + 0.5(1.50) + 0.3(0.70) = \mathbf{1.16}, \quad \hat{y}_2 = \mathbf{0.53}$$

How it works

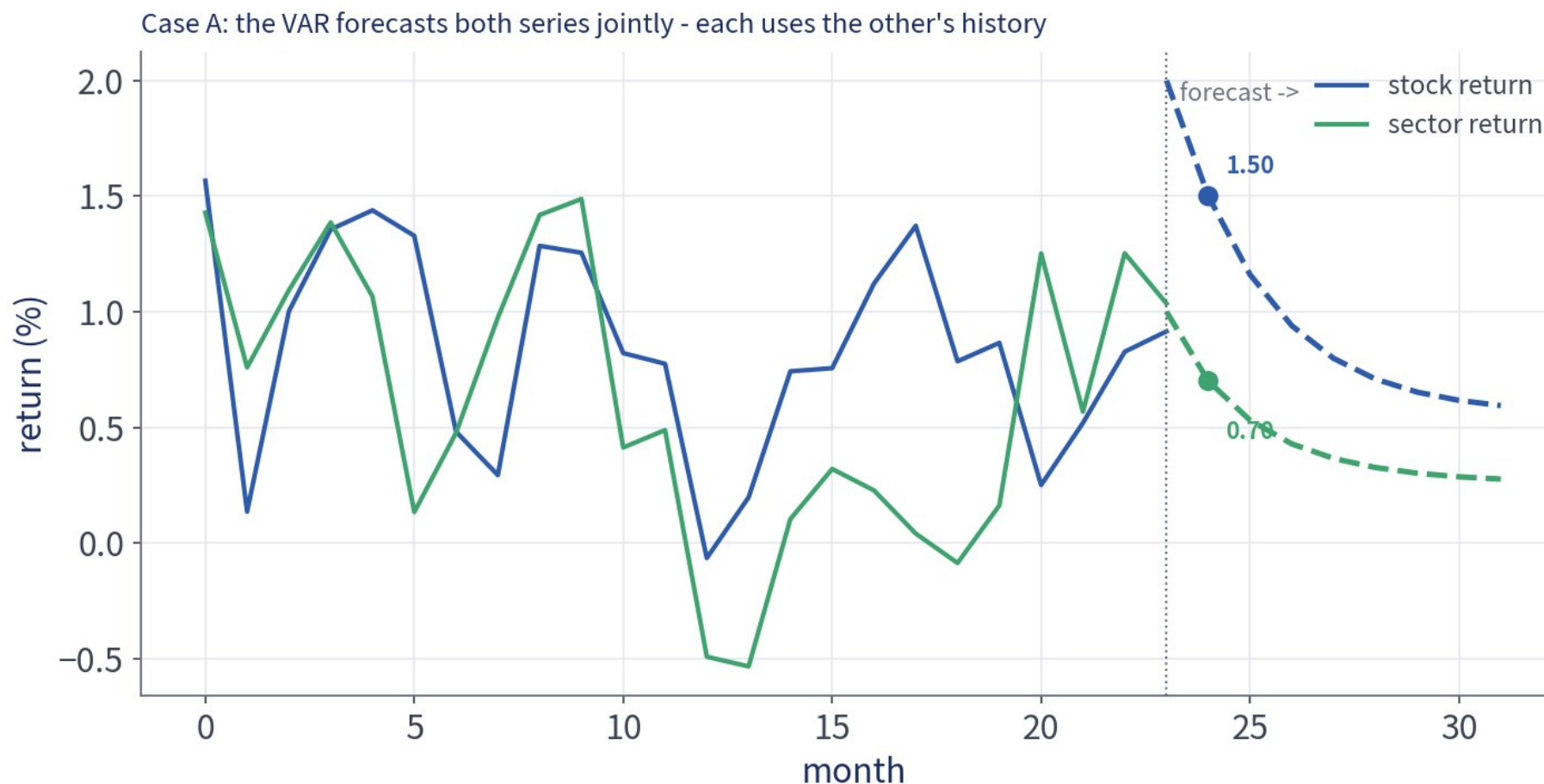
- Use the one-step forecast as the new input.
- Multiply by the matrix again.
- Add the constant.

The result

- Stock: 1.16%, sector: 0.53%.
- Both drift toward their long-run means.
- Because the VAR is stable.

Forecasting Both at Once

The stock and sector returns are modeled jointly, so each forecast draws on the other's history. Past the dotted line, the VAR projects both series forward together - and because it's stable, they settle toward their long-run means rather than wandering off



Why Joint Beats Isolated

The cross term earns its keep

What one-at-a-time misses

- The stock's move often follows the sector's.
- A univariate model ignores that lead-lag.
- It leaves information on the table.

What the VAR captures

- Feedback between the two series.
- Lead-lag through the cross terms.
- A joint forecast, internally consistent.

Stability and Lag Order

Two checks before you trust a VAR

A VAR is only useful if it doesn't explode and isn't overfit. Stability comes from the eigenvalues of the coefficient matrix: all must sit inside the unit circle. And the lag order p is chosen by information criteria - where AIC and BIC often disagree, BIC favoring the simpler model. We check both by hand

The two checks

- Stability: eigenvalues inside the unit circle.
- Lag order: minimize AIC or BIC.
- Both guard the forecast.

What we'll do by hand

- Compute the eigenvalues of the matrix.
- Confirm both are below one.
- Compare AIC and BIC across lags.

The Stability Condition

The multivariate unit-root test

stable $\iff$ all eigenvalues of A have $|\lambda| < 1$ (the multivariate unit-root condition)

The condition

- Look at the eigenvalues of the matrix.
- Each must have modulus below one.
- Then shocks fade and forecasts converge.

What failure looks like

- An eigenvalue at or above one.
- Shocks amplify instead of decay.
- Forecasts explode - the model is unusable.

The Eigenvalues

Both inside the unit circle

$$\lambda = \frac{\text{tr} \pm \sqrt{\text{tr}^2 - 4 \det}}{2} = \frac{0.9 \pm \sqrt{0.13}}{2} = \mathbf{0.63, 0.27} \text{ (both } < 1 \Rightarrow \text{stable)}$$

The calculation

- Trace 0.9, determinant 0.17.
- Eigenvalues 0.63 and 0.27.
- Both comfortably below one.

The contrast

- A matrix with entries 0.9 and 0.4 gives 1.25.
- Above one - that VAR would explode.
- Small coefficients keep it stable.

Choosing the Lag Order

Where AIC and BIC disagree

lag p	AIC	BIC	note
1	-8.20	-8.05	BIC's choice
2	-8.28	-8.02	AIC's choice
3	-8.25	-7.88	both worse

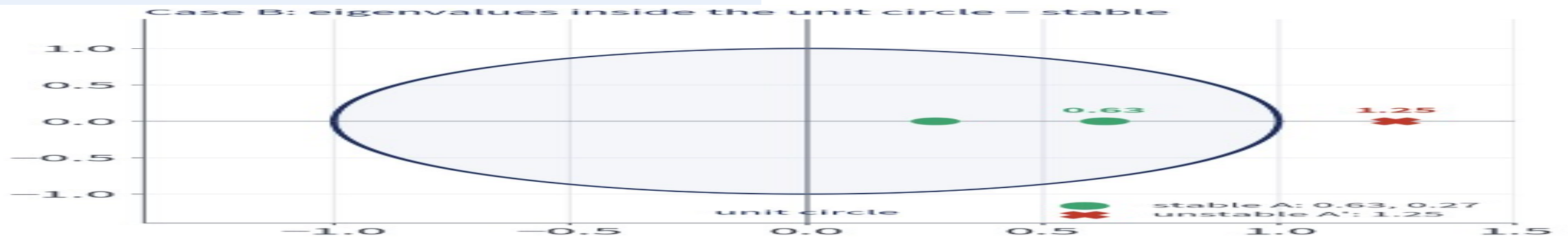
AIC picks the lowest value at $p=2$; BIC, with its heavier penalty on parameters, picks $p=1$. When they disagree, BIC's simpler model usually forecasts more reliably out of sample.

Inside the Circle Is Safe

The eigenvalues, plotted

Reading the picture

- The navy ring is the unit circle.



Every eigenvalue must land inside the ring

- for the VAR's forecasts to converge.

Guard the Forecast

Stability and parsimony together

Stability matters because

- An unstable VAR gives exploding forecasts.
- Shocks never die out.
- Always check the eigenvalues first.

Lag order matters because

- Too many lags overfit and forecast worse.
- Too few miss real dynamics.
- Let BIC lean you toward simplicity.

The VAR Routine

Check the model against the forecast

The whole workflow is short. Confirm the series belong together, pick the lag order with an information criterion, estimate each equation by ordinary least squares, and check stability through the eigenvalues. Then forecasting is just repeated matrix multiplication. The cross terms are what make it worth the trouble -

The forecast (Case A)

- One matrix multiply per step.
- Stock 1.50%, sector 0.70% next month.
- Each borrows the other's history.

The checks (Case B)

- Eigenvalues 0.63 and 0.27 - stable.
- BIC prefers $p=1$.
- Guarded and parsimonious.

Reproduce the Forecast and the Checks

Direct Claude Code to check your arithmetic

GOAL Forecast the bivariate VAR, then verify stability and lag selection.

[INPUT] Prompt · Claude Code

```
Write and run part1_practice.py that:  
CASE A - forecast:  
  * with  $c=[0.2,0.1]$ ,  $A=[[0.5,0.3],[0.1,0.4]]$ , and  
     $y=[2.0,1.0]$ , print the 1-step and 2-step forecasts.  
CASE B - stability and lag:  
  * print the eigenvalues of A and confirm  $|\lambda|<1$ ;  
  * show  $A'=[[0.9,0.4],[0.3,0.9]]$  is unstable;  
  * fit a VAR at  $p=1,2,3$  and print AIC and BIC.  
Then plot the joint forecast and the eigenvalues in  
the unit circle.
```

Check against the slides

- 1-step $[1.50, 0.70]$.
- 2-step $[1.16, 0.53]$.
- eigenvalues 0.63, 0.27.
- AIC picks $p=2$, BIC $p=1$.

Deliverable: part1_practice.py whose printed numbers match the forecasts and the checks.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 Where do a VAR's cross-dynamics live?

In the off-diagonal coefficients of the matrix

Q2 When is a VAR stable?

When every eigenvalue sits inside the unit circle

Q3 Why might you follow BIC over AIC?

It favors the simpler, more robust model

All three come from the VAR forecast, the eigenvalue check, and the lag selection.

Three Pitfalls to Avoid

Where VAR modeling trips people

Common mistakes

- "Skip the stability check and just forecast."
- "More lags always fit better."
- "Model each series on its own."

The reality

- An unstable VAR explodes - check eigenvalues first.
- Extra lags overfit; let BIC discipline p .
- The cross terms are the whole point of a VAR.

Two Calculations to Remember

Part 1 applications in one screen

1

Forecast two series with one matrix multiply

From last month's $[2.0, 1.0]$ and the coefficient matrix, the one-step forecast was $[1.50, 0.70]$ and the two-step $[1.16, 0.53]$. The cross term carried the sector's move into the stock's forecast - the payoff of modeling jointly.

2

Check stability, then pick the lag

The eigenvalues of the matrix were 0.63 and 0.27 - both inside the unit circle, so the VAR is stable and its forecasts converge. Across lags, AIC picked $p=2$ and BIC $p=1$; when they disagree, BIC's simpler model usually travels better.

3

The routine is short and repeatable

Pick the lag, estimate each equation by least squares, confirm stability, then forecast by iterating the matrix. The cross-dynamics are what make the whole exercise worthwhile.

Next — Part 2 practice: Granger causality, testing whether one series really helps predict another.

★ WEEK 6 · PART 2 of 3

Granger Causality

Does the Past of X Help Predict Y?

The predictive test · four outcomes · why it is NOT real causality

A fitted VAR lets us ask a sharp question: does one series' history improve the forecast of another? That is Granger causality — powerful, useful, and dangerously easy to misread.

2

PART 2 · GRANGER CAUSALITY

Predictive, Not Causal

The Granger test, its four possible outcomes, the crucial warning about what it does and does not mean, and its honest use for lead-lag analysis in markets.

From Co-Movement to a Question

The VAR sets up a testable relationship

1

The VAR estimated the cross-terms

Part 1 gave us off-diagonal coefficients linking each series to the other's past. Now we ask whether those links are statistically real.

2

The question is about prediction

Does knowing X's history let us forecast Y better than knowing Y's history alone? If so, X 'Granger-causes' Y.

3

The answer is easy to over-interpret

A significant test is about predictive content, not cause and effect. Getting this distinction right is the heart of Part 2.

Part 2 in one line: Granger causality tests prediction, and prediction is not causation.

The Core Question

Does X's past improve the forecast of Y?

Granger's insight: if the past of X helps predict Y beyond what Y's own past already provides, then X carries useful information about Y's future. That is all the test really claims.

The intuition

- Start with the best forecast of Y from its own history.
- Now add X's history. Does the forecast improve?
- If yes, X 'Granger-causes' Y — it leads Y in a predictive sense.

Why it is testable

- It reduces to comparing two forecasts' errors.
- That comparison is a standard statistical test.
- No hidden theory — just predictive value, measured.

Granger Causality, Defined

A statement about forecast error

$$X \text{ Granger-causes } Y \iff \text{MSE}(Y_t | Y_{<t}, X_{<t}) < \text{MSE}(Y_t | Y_{<t})$$

Reading it

- The left side conditions on both series' past.
- The right side conditions on Y's past alone.
- If adding X strictly lowers the forecast error, X Granger-causes Y.

What it is really saying

- 'Cause' here means 'helps predict', nothing more.
- It is defined entirely in terms of forecast accuracy.
- The name is historical — and, we'll see, misleading.

Restricted vs Unrestricted

An F-test on the cross-lag coefficients

$$\underbrace{Y_t = c + \sum a_i Y_{t-i} + \sum b_i X_{t-i} + \varepsilon_t}_{\text{unrestricted}} \xrightarrow{F\text{-test}} H_0 : b_1 = \dots = b_q = 0$$

The mechanics

- Fit Y on its own lags (restricted).
- Fit Y on its own lags PLUS X's lags (unrestricted).
- F-test whether X's lag coefficients are jointly zero.

The verdict

- Reject the null $\rightarrow$ X's past adds predictive value.
- Fail to reject $\rightarrow$ X adds nothing beyond Y's own past.
- It is a joint test on all of X's lag terms at once.

Does X's Past Add Anything?

Forecast Y from its own past, then again with X's past added. If the error drops significantly, X Granger-causes Y — the whole test in one picture.

The Test: Does X's Past Add Anything?

Forecast Y two ways and compare. If adding X's past cuts the error, X Granger-causes Y.

Restricted model

predict Y from its OWN past only

$$Y_t = c + a_1 Y_{(t-1)} + \dots + e_t$$

gives a forecast error:

MSE (restricted)

Unrestricted model

add the past of X as well

$$Y_t = c + a Y_{(t-1)} + b X_{(t-1)} + e_t$$

gives a smaller error?

MSE (unrestricted)

The F-test asks: is the drop in error statistically real?

If yes (the b coefficients are jointly non-zero) → X Granger-causes Y. If no → X adds no predictive value.

Running the Test

Five steps to a clean Granger result

1 · Stationarize

both series

Difference so X and Y are stationary (Part 1).

2 · Pick lags

order p

Use the VAR lag order; same p for both.

3 · Two fits

restrict?

Estimate the restricted and unrestricted models.

4 · F-test

jointly zero?

Test X's lag coefficients jointly against zero.

5 · Both ways

direction

Repeat with roles swapped to get the direction.

Always test BOTH directions — $X \rightarrow Y$ and $Y \rightarrow X$ are separate questions with separate answers.

Four Possible Outcomes

A pair relates in one of four ways

One-directional

- X Granger-causes Y but not the reverse — X leads Y.
- Or Y Granger-causes X — Y leads X.
- A clear lead-lag ordering in the predictive sense.

Two-directional or none

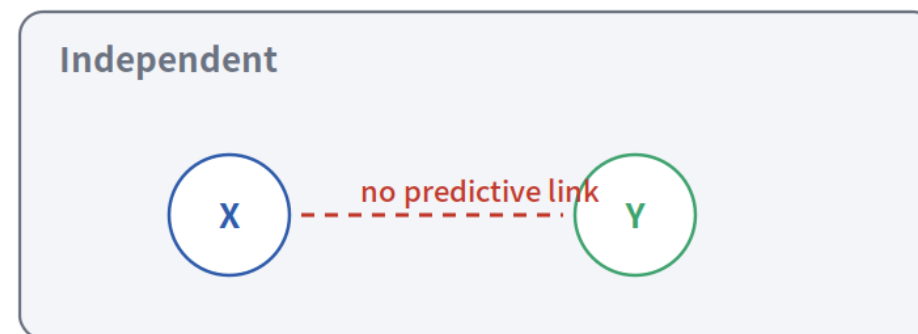
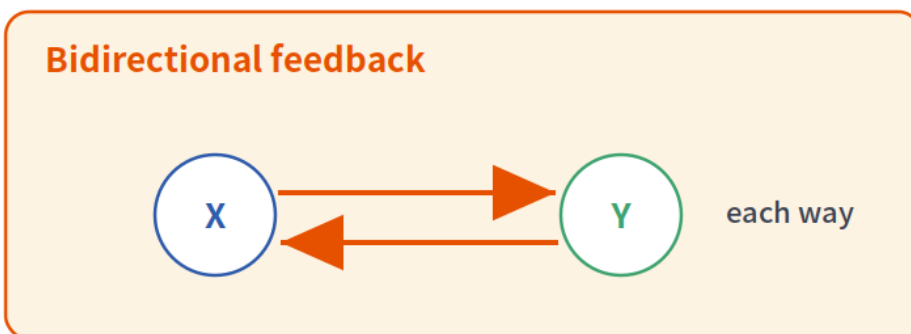
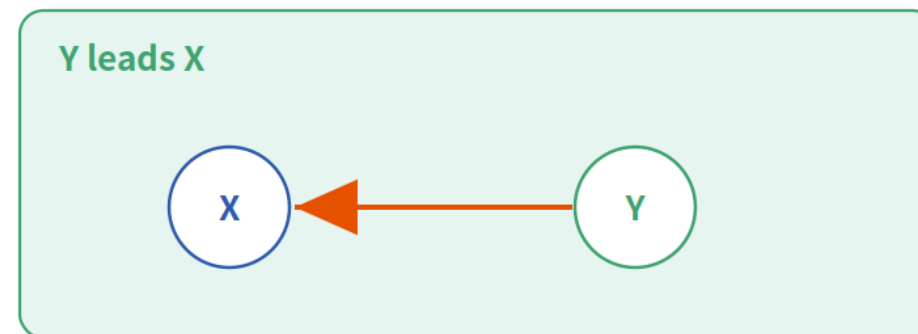
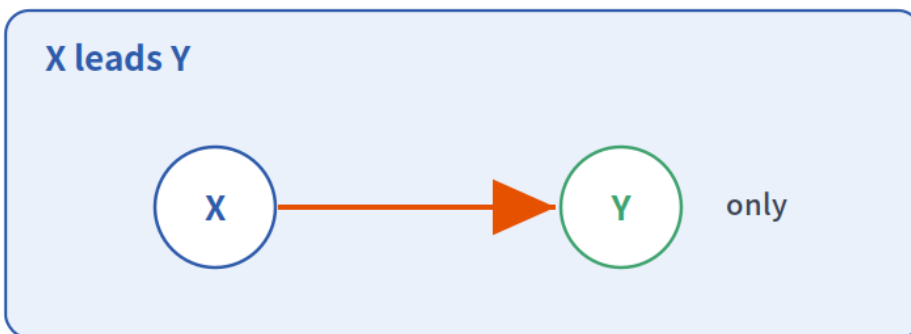
- Both directions significant — a feedback loop.
- Neither significant — no predictive link between them.
- Each case tells a different story about the pair.

Directions of Granger Causality

X leads Y, Y leads X, mutual feedback, or independence. Testing both directions places every pair into exactly one of these four boxes.

Four Possible Outcomes

A pair can relate in one of four predictive ways.



Granger Causality Is NOT Real Causality

A name that has misled generations

Granger causality means 'X's past helps predict Y'. It does NOT mean X makes Y happen. A predictive link can arise with no causal mechanism at all — the name promises far more than the test delivers.

What it DOES establish

- That X's history contains information about Y's future.
- A directional, time-ordered predictive relationship.
- A useful signal for forecasting and screening.

What it does NOT establish

- That intervening on X would change Y.
- That there is any causal mechanism at all.
- That a hidden third variable isn't driving both.

Why the Name Misleads

Three ways prediction diverges from cause

Common culprits

- A confounder Z drives both X and Y at different lags.
- Expectations: markets price the future, reversing apparent lead-lag.
- Timing artifacts from sampling or reporting delays.

The honest reading

- 'X Granger-causes Y' = 'X's past predicts Y'. Full stop.
- Add 'in a purely statistical, non-causal sense' in your head.
- Reserve causal claims for experiments or careful design.

A Cautionary Tale

When prediction has no causal content

A falling barometer 'Granger-causes' storms: its past predicts them. But smashing the barometer will not stop the weather — a hidden driver (air pressure) moves both.

The structure

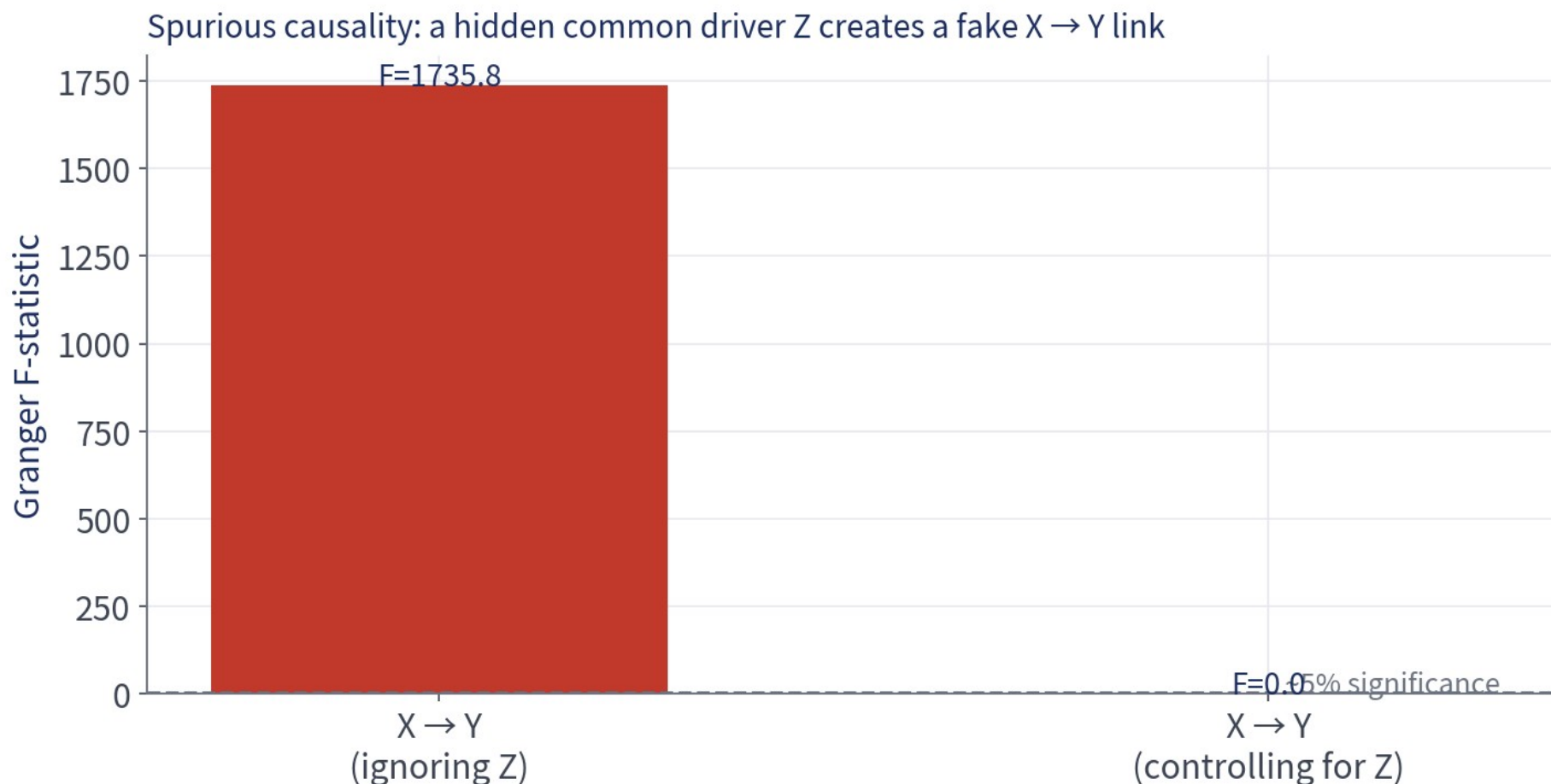
- A common cause (pressure) drives both barometer and storm.
- The barometer reacts first, so its past predicts the storm.
- The test correctly finds predictive content — and nothing more.

The finance version

- One asset may lead another only because both track a factor.
- The 'leader' is just quicker to reflect shared news.
- Trading on it as a cause can fail the moment the factor shifts.

Spurious Causality From a Confounder

X appears to strongly Granger-cause Y — until you control for the hidden common driver Z, and the effect collapses. Predictive links can be entirely indirect.



Instantaneous Causality

When two series move in the SAME period

instantaneous causality $\iff \text{Cov}(\varepsilon_{1,t}, \varepsilon_{2,t}) \neq 0$ (contemporaneous)

What it is

- Granger causality is about LAGGED prediction.
- But shocks can also be correlated within the same period.
- That contemporaneous link is 'instantaneous causality'.

Why it matters

- It shows up as correlated VAR residuals.
- It has no direction — it is symmetric by nature.
- It is exactly what forces us to orthogonalize in Part 3.

Beyond Pairs: Block Exogeneity

Granger causality in a larger system

The generalization

- With more than two variables, test a GROUP of lags jointly.
- Does a block of variables help predict another?
- This is 'block exogeneity' — the multivariate Granger test.

Why condition on others

- Testing $X \rightarrow Y$ alone can be fooled by a third variable.
- Including it can make a spurious link disappear.
- Always ask: what else should be in the system?

Lead-Lag in Markets

Where Granger causality earns its keep

Used honestly — as a forecasting and screening tool, not a causal claim — Granger causality reveals which series carry early information about others: the essence of lead-lag analysis.

Classic lead-lag patterns

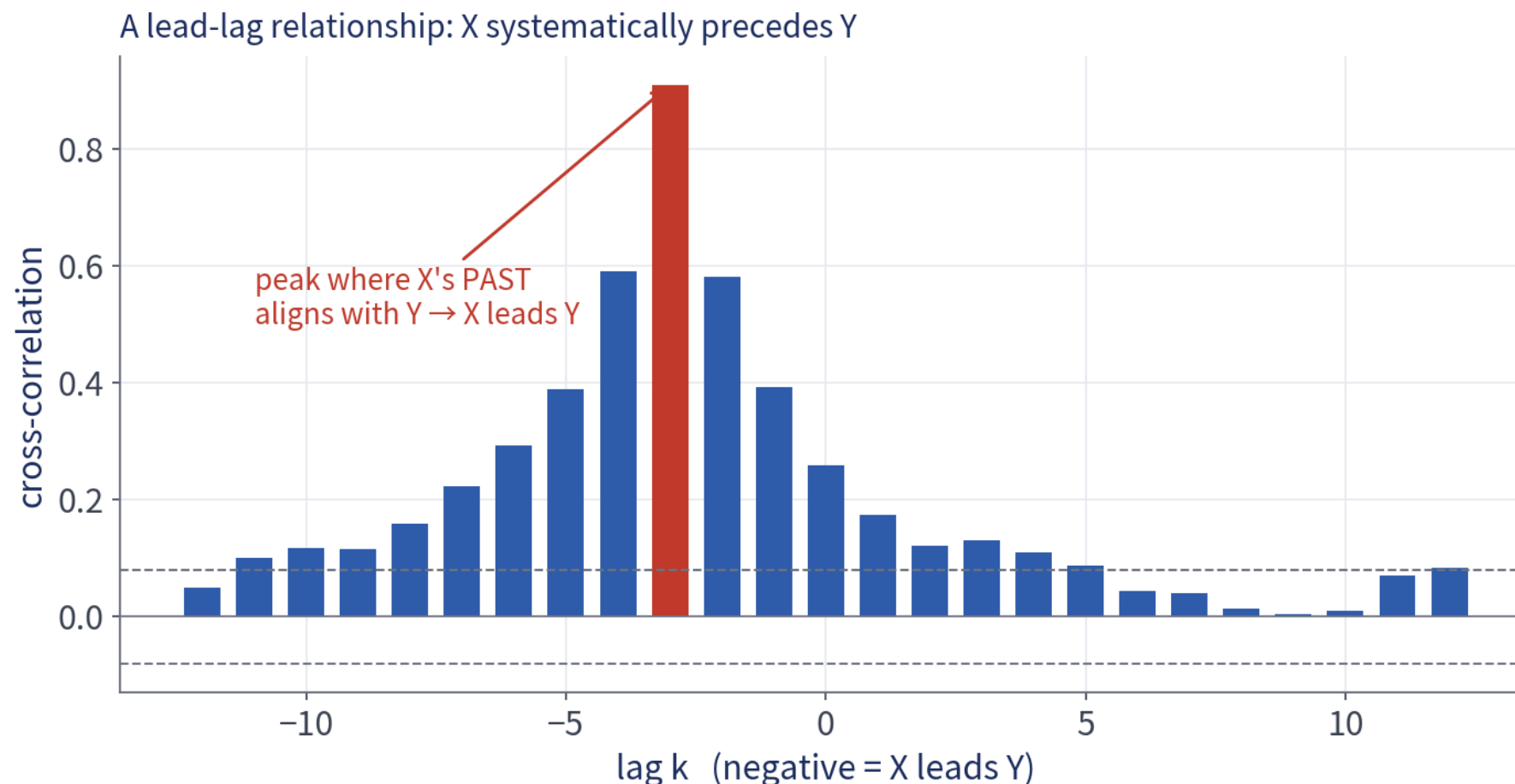
- Futures often lead spot prices.
- Large, liquid stocks can lead smaller, related ones.
- Some macro releases lead sector returns.

How to use it

- As a screen for candidate predictors, then validate out of sample.
- As a description of information flow, not a trading guarantee.
- Combine with economic reasoning — never rely on it alone.

A Lead-Lag Relationship

The cross-correlation peaks where X's past aligns with Y's present: X systematically precedes Y. That predictive lead is what a Granger test formalizes.



Correlation, Granger, Causation

Three rungs on the same ladder

Property	Correlation	Granger causality	True causation
Uses time order	no	yes	yes
Predictive	weak / static	yes, by definition	yes
Rules out confounders	no	no	yes (with design)
What it licenses	association	forecasting	intervention

Granger sits between correlation and causation: it adds time and prediction, but not the right to intervene.

What a Significant Result Tells You

And, just as important, what it doesn't

It DOES tell you

- X's past has predictive value for Y in this sample.
- There is a directional, time-ordered statistical link.
- X is a candidate input for forecasting Y.

It does NOT tell you

- That X causes Y in any mechanical sense.
- That the link will hold out of sample or persist.
- That no omitted variable explains it away.

Granger Causality and Efficient Markets

Why strong predictability is rare and fragile

The efficiency view

- If X reliably predicted Y's returns, traders would exploit it.
- Doing so tends to erase the very pattern (Week 1).
- So robust, tradable Granger causality in returns is scarce.

Where it does appear

- At high frequency, as information propagates between assets.
- In volumes, volatility, and fundamentals more than raw returns.
- As transient lead-lag that decays as it becomes known.

Using Granger Causality Well

A disciplined checklist

Do

- Treat it as a forecasting and screening tool.
- Test both directions and condition on relevant variables.
- Validate any predictor out of sample before trusting it.

Don't

- Don't read it as cause and effect.
- Don't ignore possible confounders or expectations.
- Don't assume a sample result will persist.

Test Granger Causality

Direct Claude Code through the test — and its caveats

GOAL Test Granger causality both ways on a pair, then show how a confounder can create a spurious result.

[INPUT] Prompt · Claude Code

Write and run `granger_lab.py` that:

1. takes two stationary series and runs the Granger test in BOTH directions with `statsmodels`, reporting the F-stats and p-values by lag;
2. plots their cross-correlation to show any lead-lag;
3. then simulates a confounder `Z` driving both `X` and `Y` at different lags, and shows the spurious `X→Y` result vanishing once `Z` is controlled for.

Interpret each result carefully.

How to work with it

- Report BOTH directions — they can differ.
- The CCF peak location shows who leads whom.
- Watch the spurious link die when `Z` is added.
- Never phrase the output as 'X causes Y'.

Deliverable: `granger_lab.py` + a both-directions result and a confounder demonstration.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1	The test says X Granger-causes Y. Can you conclude X causes Y?	No — prediction only
Q2	Both $X \rightarrow Y$ and $Y \rightarrow X$ are significant. What is this called?	Bidirectional feedback
Q3	A significant link vanishes when you add variable Z. What happened?	Z was a confounder

All three come from the meaning of the test and the confounder problem.

Three Misconceptions to Avoid

The errors this whole part guards against

Common myths

- "Granger causality proves X causes Y."
- "One direction is enough to characterize a pair."
- "A significant in-sample result will make money."

The reality

- It proves predictive content only — never a mechanism.
- Test both directions; feedback and independence are distinct.
- Validate out of sample; efficient markets erase easy edges.

Three Things to Carry Forward

Part 2 in one screen

1

Granger causality is a prediction test

It asks whether X's past improves the forecast of Y — an F-test comparing a restricted and an unrestricted model, run in both directions.

2

It is not real causality

Confounders, expectations, and timing can create predictive links with no mechanism. Read every result as 'helps predict', never 'causes'.

3

Used honestly, it maps information flow

As a screening and lead-lag tool — validated out of sample and conditioned on the right variables — it reveals which series carry early information.

Next — Part 3: correlated shocks, orthogonalization, and tracing shocks with IRF and FEVD.

★ WEEK 6 · PART 2 · PRACTICE

Granger Causality, Worked by Hand

Does One Series Help Forecast Another - and Which Way?

The F-test on cross-lags, and the four outcomes

Part 2's test, with the arithmetic shown. We run the Granger F-test - does adding X's past lags cut the forecast error for Y? - then test the reverse direction to classify the pair, and confront why a significant result is prediction, not proof of cause.

2P

PART 2 · APPLICATIONS

Prediction, Not Proof

Case A: the Granger F-test - fit Y on its own past, then add X's lags, and test whether the error drop is significant.

Case B: test the reverse direction to place the pair among the four outcomes, and see why Granger causality isn't real causality. Then a Claude Code lab.

What Part 2 Told Us

Now we test it - with numbers

1

Granger causality is about forecasts

X Granger-causes Y if X's past improves the forecast of Y beyond Y's own past - a statement about prediction, not mechanism.

2

It's an F-test on cross-lag coefficients

Fit Y restricted, then unrestricted with X's lags added. If the residual sum of squares drops enough, X helps.

3

A significant result is not real causality

Prediction can come from a common driver or anticipation. Granger causality answers a narrower, forecasting question.

Everything here is a calculator exercise: one F-test, one reverse test, one classification.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a restricted and unrestricted fit.
- Compute the F, or test the reverse direction.
- Then read the verdict - carefully.

What you should watch

- How the RSS drop becomes an F-statistic.
- How the reverse test sets the direction.
- Why prediction is not the same as cause.

The Granger F-Test

Does X help forecast Y?

We ask whether a sector index's past returns help forecast a stock's return, beyond the stock's own history. Fit the stock on its own lags - that's the restricted model - then add the sector's lags for the unrestricted one. If the residual error drops enough, an F-test declares the sector Granger-causes the stock. We compute it by

The setup

- Y is the stock, X the sector index.
- Two lags each, 100 observations.
- Restricted RSS 52, unrestricted 45.

What we'll do by hand

- Form the F from the RSS drop.
- Compare to the F critical value.
- Decide whether X helps.

Restricted vs Unrestricted

The forecast error, with and without X

$$\text{restricted: } Y_t = \sum a_i Y_{t-i} \quad \text{vs} \quad \text{unrestricted: } Y_t = \sum a_i Y_{t-i} + \sum b_i X_{t-i} \quad (\text{test } b_i=0)$$

The two models

- Restricted: Y on its own past only.
- Unrestricted: add X's past lags.
- The test asks if those lags matter.

The null

- All of X's lag coefficients are zero.
- Then X adds nothing to the forecast.
- The F-test tries to reject that.

The F-Statistic

From the drop in residual error

$$F = \frac{(\text{RSS}_r - \text{RSS}_u)/q}{\text{RSS}_u/(T-k)} = \frac{(52-45)/2}{45/95} = \frac{3.5}{0.474} = \mathbf{7.39}$$

The numerator

- The RSS falls from 52 to 45.
- A drop of 7, over 2 restrictions.
- That's 3.5 per restriction.

The denominator

- The unrestricted RSS over its degrees of freedom.
- 45 over 95, about 0.47.
- The ratio is an F of 7.39.

The Verdict

Against the F critical value

$$7.39 > \underbrace{3.09}_{F(2,95) \ 5\%} \Rightarrow \text{reject } b_i=0 \Rightarrow X \text{ Granger-causes } Y \text{ (its past adds forecast power)}$$

The F of 7.39 far exceeds the 5% critical value of 3.09, so we reject the null that X's lags are all zero. The sector's past does improve the stock's forecast - X Granger-causes Y.

The Error Drop, and the F

Adding X's lags cuts the residual sum of squares from 52 to 45 - a drop of 7. Scaled into an F-statistic, that gain reaches 7.39, well above the 3.09 bar. The reduction is large enough that random chance is an implausible explanation. X's past genuinely helps

Case A: X's past lags cut the forecast error enough to matter



What 'Helps Forecast' Means

A prediction claim, precisely

What it does say

- X's past cuts Y's forecast error.
- The reduction is statistically real.
- X carries predictive information about Y.

What it doesn't say

- That X mechanically drives Y.
- That acting on X will change Y.
- Only that X predicts Y - for now.

Direction, and the Caveat

Which way, and what it really means

Granger causality has a direction, so we test the reverse: does the stock's past help forecast the sector? Running the same F-test the other way gives a small, insignificant statistic - so the relationship is one-directional. That places the pair in one of four outcomes. But even a strong result is prediction, not proof of cause

The reverse test

- Now forecast X from its own past.
- Add Y's lags and test them.
- The RSS barely moves.

What we'll do by hand

- Compute the reverse F.
- Classify among the four outcomes.
- Then weigh the causality caveat.

The Four Outcomes

Every pair falls into one

four outcomes: $X \rightarrow Y$ only | $Y \rightarrow X$ only | both (feedback) | neither

How to place a pair

- Test both directions with the F.
- Each direction is significant or not.
- The two answers pick the outcome.

The four cells

- One-way each direction, feedback, or none.
- Feedback means both tests reject.
- Neither means both fail to reject.

Test the Reverse

Does Y help forecast X?

$$\text{reverse } Y \rightarrow X : F = \frac{(60-58.5)/2}{58.5/95} = \mathbf{1.22} < 3.09 \Rightarrow \text{not significant} \Rightarrow X \rightarrow Y \text{ only}$$

Forecasting the sector from its own past, adding the stock's lags barely lowers the RSS - the reverse F is just 1.22, well below 3.09. So the stock does not Granger-cause the sector: the relationship runs one way, X to Y.

Classify, Then Beware

One-directional - but not causal proof

The classification

- X to Y: significant ($F = 7.39$).
- Y to X: not significant ($F = 1.22$).
- Outcome: X Granger-causes Y only.

The caveat

- A hidden common driver could fake this.
- Or the stock anticipates the sector's news.
- Prediction is not proof of mechanism.

Placing the Pair

The four-outcome grid

Reading the grid

- Columns: does X Granger-

X Granger-causes Y?
Case B: our pair is X → Y only (F 7.39 yes, reverse 1.22 no)

Y → X?	YES	X → Y only	both (feedback)
	NO	neither	Y → X only
		YES	NO (rev)

- So it lands in X to Y only (green).

Why the Name Misleads

And why it matters in markets

Three ways it isn't cause

- A common factor drives both series.
- One series anticipates the other's news.
- The lead-lag reverses out of sample.

The efficient-markets lens

- Strong return predictability is rare.
- If it were easy, it would be arbitrated away.
- Findings are often fragile - treat with care.

The Granger Routine

Five steps, one honest conclusion

The test is a fixed routine. Fit Y on its own lags, add X's lags, and take the drop in residual error as an F. Compare to the critical value, then test the reverse to fix the direction. What you get is a precise forecasting statement - X's past helps predict Y - not a claim about mechanism. Used with that discipline it's a sharp tool

The test (Case A)

- RSS drop 52 to 45, $F = 7.39$.
- Clears the 3.09 bar.
- X Granger-causes Y.

The direction (Case B)

- Reverse F only 1.22.
- One-directional: X to Y.
- Prediction, not proven cause.

Reproduce the Test Both Ways

Direct Claude Code to check your arithmetic

GOAL Run the Granger F-test in both directions and classify the pair.

[INPUT] Prompt · Claude Code

```
Write and run part2_practice.py that:  
CASE A - X to Y:  
  * with RSS_r=52, RSS_u=45, T=100, k=5, q=2,  
    compute the F-statistic and the 5% critical value;  
  * decide whether X Granger-causes Y.  
CASE B - reverse and classify:  
  * with RSS_r=60, RSS_u=58.5, compute the reverse F;  
  * classify the pair among the four outcomes.  
Optional: use statsmodels grangercausalitytests on a  
simulated pair and confirm the direction.
```

Check against the slides

- $F(X \rightarrow Y) = 7.39 > 3.09$.
- $F(Y \rightarrow X) = 1.22 < 3.09$.
- critical $F(2,95) = 3.09$.
- outcome: $X \rightarrow Y$ only.

Deliverable: part2_practice.py whose printed numbers match both F-tests and the classification.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What does the Granger F-test compare?

**Restricted vs
unrestricted forecast
error**

Q2 How do you find the direction of causality?

**Test both ways; each
F is significant or not**

Q3 Why isn't Granger causality real causality?

**Prediction can come
from a confounder or
anticipation**

All three come from the F-test, the reverse test, and the causality caveat.

Three Pitfalls to Avoid

Where Granger causality trips people

Common mistakes

- "Granger causality proves X drives Y."
- "Test only one direction and stop."
- "A significant result will hold out of sample."

The reality

- It's a forecasting claim, not a mechanism.
- Always test both ways to fix the direction.
- Predictability in markets is often fragile.

Two Calculations to Remember

Part 2 applications in one screen

1

The F-test measures the forecast gain

Adding the sector's lags cut the stock's residual error from 52 to 45; scaled into an F-statistic that reached 7.39, well past the 3.09 cutoff. The sector's past genuinely improves the stock's forecast - X Granger-causes Y.

2

Test both ways, then stay humble

The reverse F was only 1.22, so the stock does not help forecast the sector: the relationship is one-directional, X to Y. But a common driver or anticipation could produce the same pattern - it's prediction, not proven cause.

3

A sharp tool, used with discipline

Fit restricted, add lags, take the error drop as an F, and check both directions. The result is an honest forecasting statement - which in efficient markets is rare and often fragile.

Next — Part 3 practice: impulse-response functions and variance decomposition, tracing shocks through the system.

★ WEEK 6 · PART 3 of 3

Shocks and Structure

Impulse Responses and Variance Decomposition

Correlated shocks · orthogonalization · IRF · FEVD · the bridge to Week 7

A VAR's real power is dynamic analysis: tracing how a single shock ripples through the whole system, and accounting for where each variable's uncertainty comes from.

3

PART 3 · SHOCKS & STRUCTURE

Tracing Shocks Through the System

Why shocks must be orthogonalized before we interpret them, how impulse responses trace their ripples, and how variance decomposition attributes uncertainty to its sources.

From Prediction to Propagation

The last, and most powerful, use of a VAR

1

Part 1 built the model, Part 2 tested links

We can fit a VAR, forecast jointly, and ask who predicts whom. Now we ask a deeper question: how does a shock move through the whole system over time?

2

But the shocks are tangled

A VAR's residuals are correlated — you cannot move one without the others. We must first untangle them into independent 'structural' shocks.

3

Then two tools finish the job

Impulse responses trace a shock's ripples; variance decomposition attributes each variable's uncertainty to its sources. Together they reveal the system's dynamics.

Part 3 in one line: untangle the shocks, then trace them (IRF) and budget them (FEVD).

A Subtlety: Correlated Shocks

Why we must 'orthogonalize' before interpreting

A VAR's residuals move together — a shock to one equation typically comes with a shock to the other. So 'shock only variable 1' is not something the raw model can even represent.

What goes wrong

- The residual covariance is not diagonal — shocks are correlated.
- Shocking Y1 alone ignores its usual companion move in Y2.
- Any 'response to a Y1 shock' is then ambiguous.

What we need

- A set of shocks that are mutually uncorrelated.
- Then each can be moved on its own, cleanly.
- That transformation is orthogonalization.

Why You Can't Just Shock One Variable

Correlation ties the shocks together

The raw residuals

- In real systems, a shock to one series rarely arrives alone.
- Interest-rate and equity shocks, say, tend to co-occur.
- The residual covariance captures exactly this co-movement.

The consequence

- An 'impulse' to one raw residual implies a partner impulse.
- So the naive response mixes two shocks together.
- We need independent shocks to get a clean answer.

Orthogonalization

Turn correlated shocks into independent ones

$$\boldsymbol{\varepsilon}_t = P \mathbf{u}_t, \quad \text{Cov}(\mathbf{u}_t) = I \quad \Rightarrow \quad \text{shocks } \mathbf{u}_t \text{ are uncorrelated}$$

The idea

- Write the correlated residuals as a mix of independent shocks.
- The structural shocks $\mathbf{u}_t$ are uncorrelated, with unit variance.
- Now each $\mathbf{u}$ can be moved on its own — a clean impulse.

Why it works

- Independence means shocking one leaves the others alone.
- The matrix P encodes how they combine into observed moves.
- The standard choice for P is the Cholesky factor.

The Cholesky Decomposition

A triangular square root of the covariance

$$\Sigma = P P^{\top} \quad P \text{ lower-triangular} \quad (\text{Cholesky decomposition})$$

What it gives

- A lower-triangular P with $\Sigma = P \cdot P^{\top}$.
- It maps the independent shocks u to the observed residuals.
- Triangular means an implicit ORDERING of the variables.

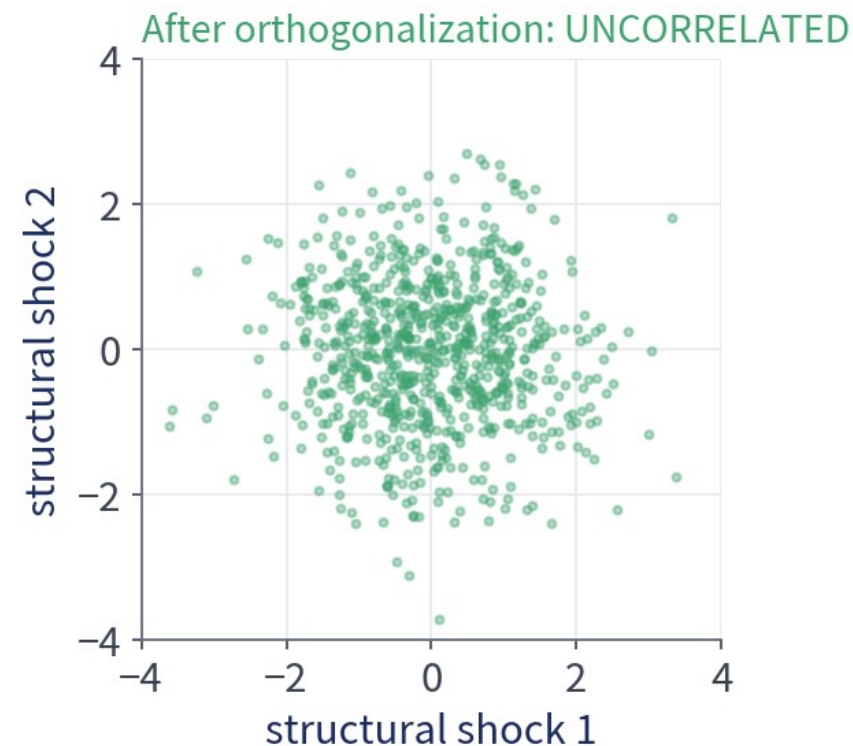
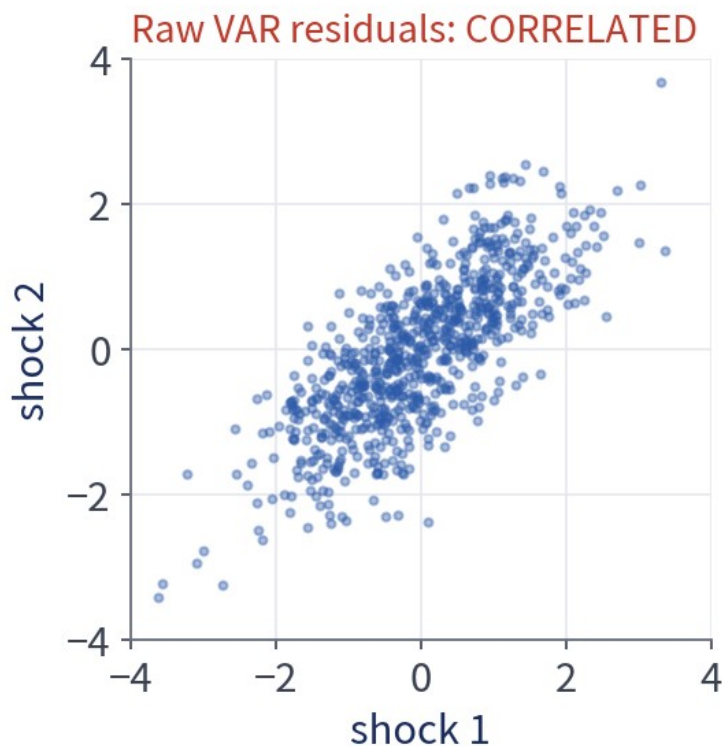
The catch

- The first variable gets all its own shock immediately.
- Later variables can respond within the period, but not vice versa.
- So the order you list variables in matters — a lot.

Untangling the Shocks

Raw VAR residuals form a tilted, correlated cloud (left). Orthogonalization rotates them into independent structural shocks (right) that can be moved one at a time.

Orthogonalization untangles correlated shocks into independent ones we can shock separately



The Ordering Problem

A choice that changes the answer

Cholesky orthogonalization assumes a causal ordering: the first variable can affect all others contemporaneously, but not the reverse. Different orderings give different impulse responses.

Why it matters

- The order encodes which shocks are allowed to hit first.
- Swap the order and the short-run responses can change.
- It is an assumption, not something the data decides.

How to handle it

- Order variables by economic reasoning (most exogenous first).
- Report sensitivity to the ordering, not a single result.
- For robust work, consider structural identification schemes.

Impulse Response Functions

How a one-time shock ripples through the system

An impulse response traces the effect of a single, isolated shock to one variable on every variable, over time. It is the clearest picture we have of a system's dynamics.

What it does

- Hit the system with one unit of a structural shock.
- Then watch every variable respond over the following periods.
- The path of each response is its impulse response function.

Why it is powerful

- It turns a matrix of coefficients into an intuitive story.
- You see the sign, the size, and how long effects last.
- It is the standard way to report VAR results.

The IRF, Defined

The response of i to shock j at horizon h

$$\text{IRF}_{ij}(h) = \frac{\partial y_{i,t+h}}{\partial u_{j,t}} = [A^h P]_{ij} \quad (\text{response of } i \text{ to shock } j \text{ at horizon } h)$$

Reading it

- Fix a shock to variable j at time t .
- The IRF is how variable i moves h periods later.
- It is computed from powers of A , scaled by the P matrix.

The shape it takes

- For a stable VAR, responses eventually decay to zero.
- They may overshoot, oscillate, or fade smoothly.
- The decay rate reflects the system's persistence.

Reading an IRF Plot

A grid of little stories

What each panel shows

- Each panel: response of one variable to one shock.
- The horizontal axis is the horizon after the shock.
- The vertical axis is the size of the response.

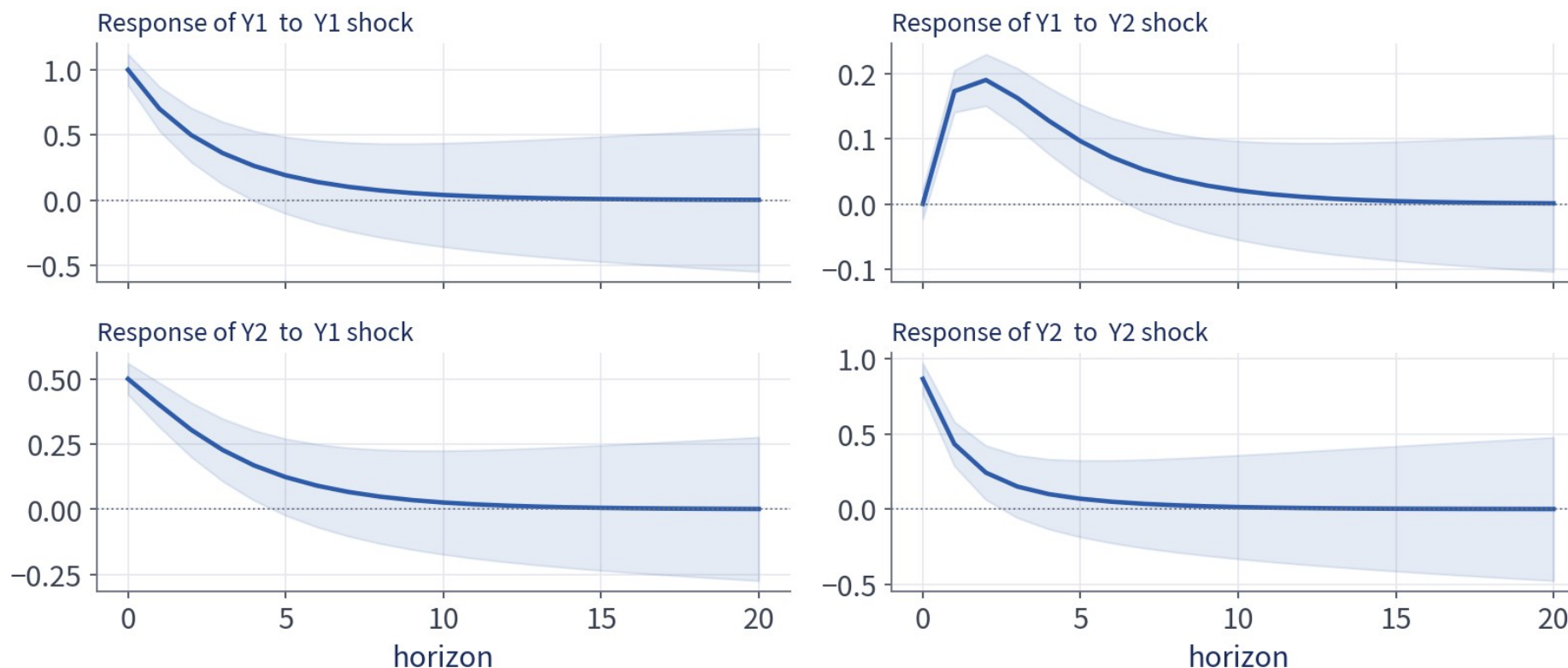
What to look for

- Sign: does the shock push the variable up or down?
- Persistence: how many periods until it fades?
- Bands: is the response statistically distinguishable from zero?

Tracing a Shock Through the System

Each panel traces one variable's response to one shock over the horizon, with uncertainty bands. Cross-panel responses show how the two series pass shocks to each other.

Impulse responses: how a one-time shock to each variable ripples through the system



What IRFs Reveal — and Their Bands

Reading the dynamics responsibly

The dynamics

- How strongly one series reacts to another's shock.
- Whether effects build, peak, and then decay.
- How long a disturbance keeps mattering.

The uncertainty

- IRFs are estimates — always plot confidence bands.
- A response is meaningful only if its band excludes zero.
- Bands usually widen at longer horizons.

Forecast Error Variance Decomposition

Where does a variable's uncertainty come from?

FEVD asks a budgeting question: of all the uncertainty in forecasting a variable, what fraction traces back to each shock? It turns the system into a variance budget by source.

The question

- Forecast a variable h steps ahead; it has some error variance.
- How much of that variance is due to its own shocks?
- How much to shocks from the other variables?

Why it complements IRF

- IRF shows the SHAPE of a shock's effect over time.
- FEVD shows the SHARE of variance each shock explains.
- Together: how it moves, and how much it matters.

The FEVD, Defined

A share of forecast-error variance

$$\text{FEVD}_{ij}(h) = \frac{\text{variance of } y_i \text{'s } h\text{-step error from shock } j}{\text{total } h\text{-step forecast error variance of } y_i}$$

Reading it

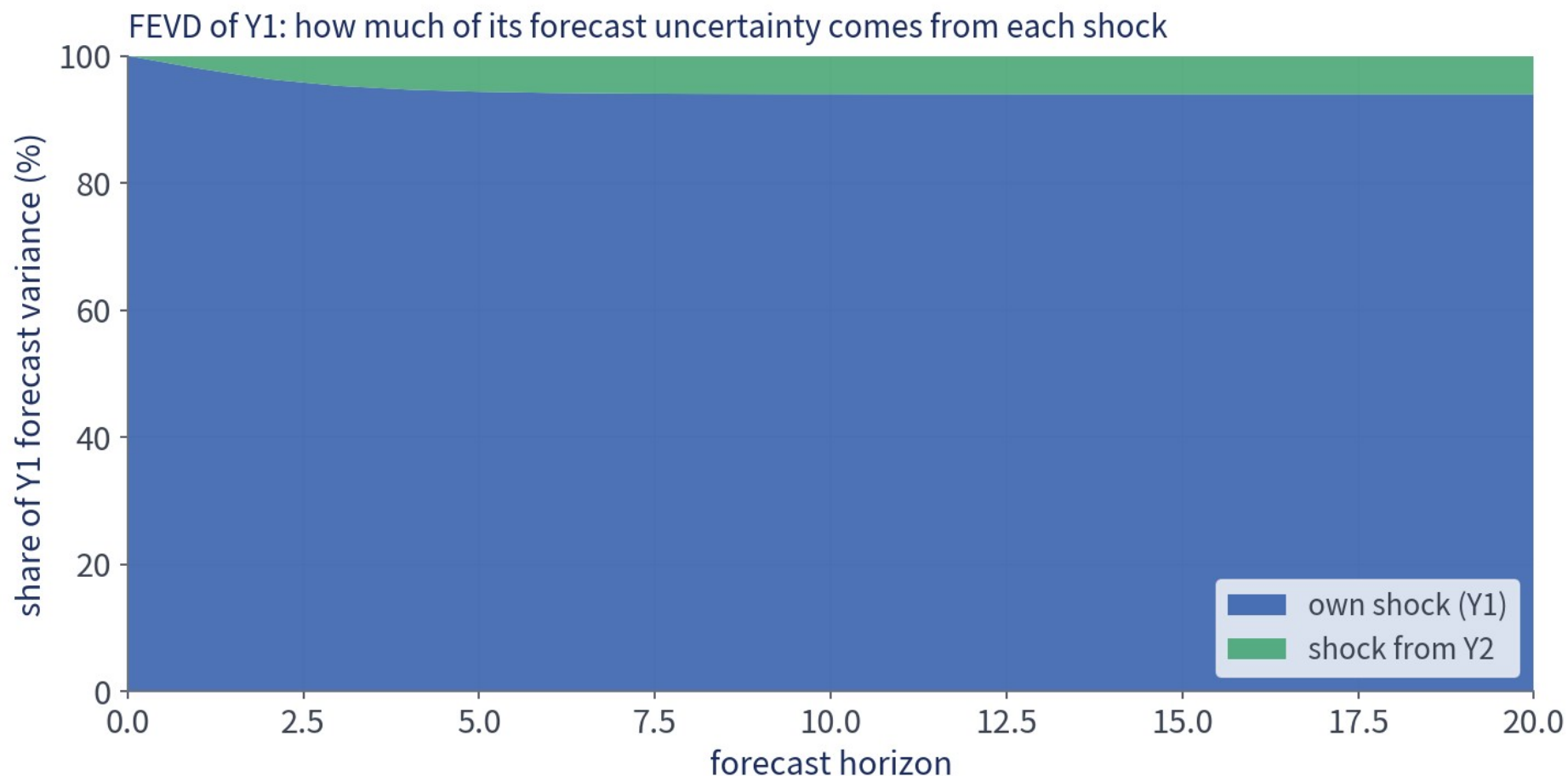
- The numerator: variance from shock j alone.
- The denominator: the variable's total forecast error variance.
- The ratio is a percentage, at each horizon h .

Its behavior

- At short horizons, a variable's own shock usually dominates.
- At longer horizons, other shocks' shares often grow.
- The shares always add up to 100% across shocks.

Where Forecast Uncertainty Comes From

For Y1, the share of forecast-error variance from its own shock versus the shock from Y2, across horizons. Early on, own shocks dominate; the other's share grows with horizon.



Reading a FEVD

A variance budget, horizon by horizon

What it tells you

- Which shocks drive a variable's uncertainty, and when.
- Whether a series is mostly self-driven or externally driven.
- How that balance shifts as the horizon lengthens.

How to use it

- Rank the sources of risk for each variable.
- Identify which series a shock elsewhere really matters for.
- Like IRF, it depends on the ordering — report sensitivity.

Granger, IRF, and FEVD

Three questions, one fitted VAR

Tool	Question it answers	What it outputs
Granger causality	Does X's past help predict Y?	yes/no, with a direction
Impulse response	How does a shock to X move Y over time?	a response path
Variance decomposition	What share of Y's uncertainty is due to X?	percentages, by horizon

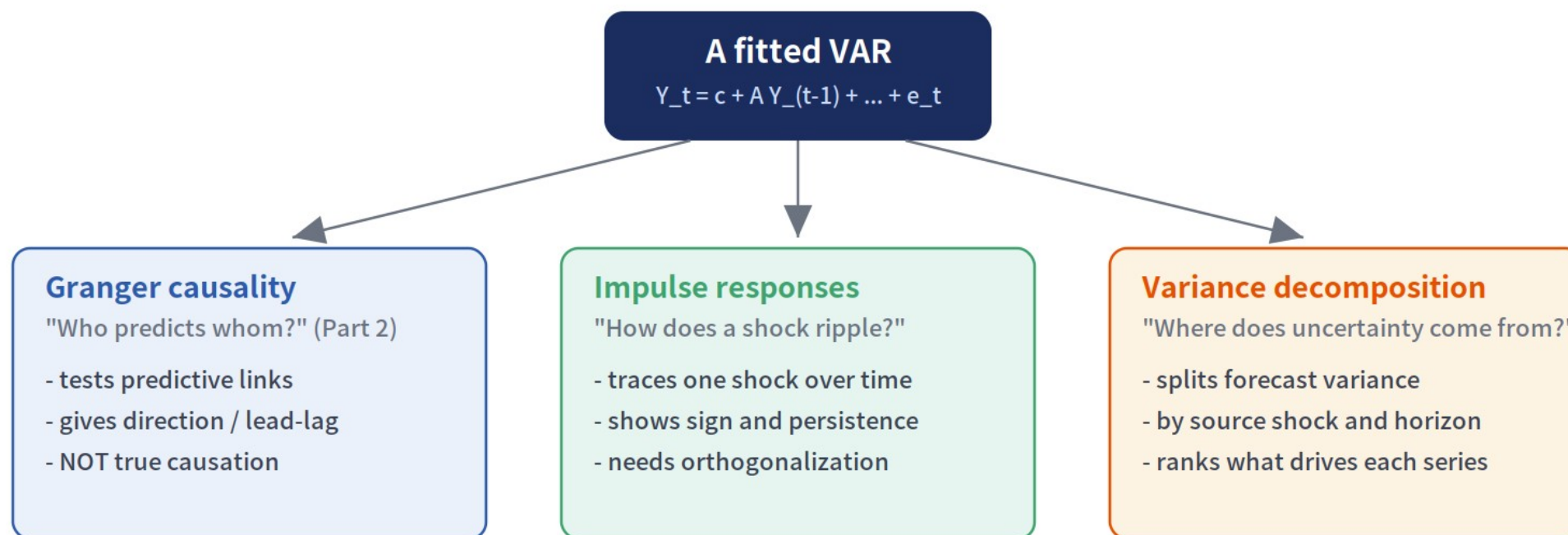
A test, a dynamic path, and a variance budget — three complementary views of the same system.

The VAR Analysis Toolkit

One fitted VAR answers three questions: who predicts whom (Granger), how shocks ripple (IRF), and where uncertainty originates (FEVD) — each a lens on the joint dynamics.

The VAR Analysis Toolkit

One fitted VAR answers three different questions - each a lens on the same joint dynamics.



The Complete VAR Picture

From data to forecasts to structure

What a VAR gives you

- Joint forecasts that respect cross-dynamics (Part 1).
- Directional predictive links via Granger tests (Part 2).
- Shock propagation and variance budgets via IRF and FEVD.

The honest caveats

- Reduced-form VARs describe, they do not prove causation.
- Structural claims (IRF, FEVD) hinge on the ordering.
- Keep the system small, stationary, and well-diagnosed.

From Co-Movement to a Tradable Anchor

How VAR sets up the next chapter

VAR models how series move together in the SHORT run. But some pairs share a LONG-run equilibrium they always return to — and that stable relationship is the anchor a pairs trade needs.

What VAR gave us

- A framework for joint dynamics and shock propagation.
- Tools to see which series lead, and how shocks spread.
- But standard VAR needs stationary (differenced) series.

Where Week 7 goes

- Cointegrated pairs share a long-run equilibrium in LEVELS.
- The error-correction model captures how they revert to it.
- That is the true statistical engine of pairs trading.

Impulse Responses and FEVD

Direct Claude Code to trace and budget shocks

GOAL From a fitted VAR, produce orthogonalized impulse responses and a variance decomposition, and probe the ordering.

[INPUT] Prompt · Claude Code

Write and run `irf_fevd_lab.py` that, from a fitted VAR on 2-3 series:

1. computes orthogonalized (Cholesky) impulse responses with confidence bands and plots the full IRF grid;
2. computes and stack-plots the FEVD for each variable across horizons;
3. RE-RUNS both with the variable order reversed and shows how much the results change.

Summarize what propagates to what, and how sensitive it is to the ordering.

How to work with it

- Read sign and persistence off each IRF panel.
- Check which shock dominates each FEVD.
- Note how reversing the order shifts short-run responses.
- This closes Week 6 and points to Week 7.

Deliverable: `irf_fevd_lab.py` + an IRF grid, a FEVD, and an ordering-sensitivity check.

Three Questions Before You Finish

If you can answer these, Week 6 has landed

Q1

Why must VAR shocks be orthogonalized before computing impulse responses?

Residuals are correlated

Q2

You reverse the variable order and the IRFs change. Why?

Cholesky ordering matters

Q3

A FEVD says Y2's shock explains 60% of Y1's variance. What does that mean?

Y2 drives Y1's uncertainty

All three come from correlated shocks, the ordering assumption, and the variance budget.

Three Pitfalls to Avoid

Where structural VAR analysis goes wrong

Common mistakes

- "I can shock one raw residual and read the response."
- "The impulse responses are the truth, whatever the order."
- "IRF and FEVD prove causation."

The reality

- Raw residuals are correlated — orthogonalize first.
- Cholesky results depend on the ordering — report sensitivity.
- They describe a reduced-form system, not proven mechanisms.

Week 6 in Three Ideas

Pulling the whole week together

1

A VAR models series jointly

Part 1: every series depends on the recent past of all series. The coefficient matrix carries the cross-dynamics, subject to stability and a parameter budget.

2

Granger tests prediction, not cause

Part 2: the F-test asks whether one series' past improves another's forecast — a directional predictive link, never a proof of causation.

3

IRF and FEVD reveal the structure

Part 3: orthogonalize the shocks, then trace their ripples (IRF) and budget the variance (FEVD) — the clearest view of the system's dynamics.

Next — Week 7: Cointegration and Error Correction, the long-run link that anchors pairs trading.

★ WEEK 6 · PART 3 · PRACTICE

IRF and FEVD, Worked by Hand

Trace a Shock Through the System, Then Budget the Variance

Cholesky, the impulse response, and the variance decomposition

Part 3's tools, with the arithmetic shown. We orthogonalize correlated shocks with a Cholesky factor and trace an impulse response through the VAR, then decompose a variable's forecast-error variance into shares - all while watching how the variable ordering changes the answer.

3P

PART 3 · APPLICATIONS

Shocks, Responses, and Variance

Case A: orthogonalize the correlated shocks with a Cholesky factor, then trace the impulse response across horizons. Case B: decompose the forecast-error variance into shares, and see how the variable ordering changes both. Then a Claude Code lab.

What Part 3 Told Us

Now we trace it - with numbers

1

Shocks are correlated, so orthogonalize

You can't shock one variable alone when shocks move together. A Cholesky factor turns them into independent unit shocks.

2

The IRF traces a shock through time

It's the response of each variable to an orthogonalized shock at every horizon - the system's dynamic story, decaying if stable.

3

The FEVD budgets the uncertainty

It splits each variable's forecast-error variance into shares from each shock - and both it and the IRF depend on the ordering.

Everything here is a calculator exercise: one Cholesky factor, one impulse response, one variance budget.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from the shock covariance and matrix.
- Trace the response, or split the variance.
- Then read the dynamics.

What you should watch

- How the Cholesky untangles the shocks.
- How the response decays over horizons.
- How ordering changes both answers.

Orthogonalize, Then Trace

A shock's path through the system

A stock's and its sector's shocks move together, so a shock to one already implies a move in the other. To interpret a clean shock, we orthogonalize with a Cholesky factor, then push a unit shock through the VAR to trace the impulse response over months. We compute the factor and the first few horizons by hand

The setup

- Shock covariance with correlation 0.4.
- The VAR matrix from Part 1.
- A unit shock to variable 1.

What we'll do by hand

- Compute the Cholesky factor.
- Read the impact response.
- Iterate the matrix for later horizons.

Orthogonalization

Untangle the correlated shocks

$$\Sigma = P P^{\top} \text{ (lower-triangular } P) : \text{ correlated shocks } \longrightarrow \text{ independent unit shocks}$$

The problem

- Shocks are correlated, so they co-move.
- Shocking one alone isn't well-defined.
- Interpretation needs independent shocks.

The fix

- Factor the covariance as P times P -transpose.
- P is lower-triangular - the Cholesky.
- Its columns are independent unit shocks.

The Cholesky Factor

A triangular square root

$$\Sigma = \begin{bmatrix} 1.0 & 0.4 \\ 0.4 & 1.0 \end{bmatrix} \Rightarrow P = \begin{bmatrix} 1.0 & 0 \\ 0.4 & 0.917 \end{bmatrix} \quad (P_{22} = \sqrt{1 - 0.4^2})$$

The entries

- Top-left is the square root of 1.0.
- Bottom-left is 0.4 - the correlation.
- Bottom-right is root of one minus 0.16.

What it means

- A shock to var 1 hits var 1 by 1.0.
- And var 2 by 0.4 - the shared part.
- That's the orthogonalized impact.

The Impulse Response

Push the shock through the matrix

$$\text{IRF}(h) = A^h P e_1 : \begin{bmatrix} 1.00 \\ 0.40 \end{bmatrix} \rightarrow \begin{bmatrix} 0.62 \\ 0.26 \end{bmatrix} \rightarrow \begin{bmatrix} 0.39 \\ 0.17 \end{bmatrix} \text{ (decays)}$$

The impact response is $[1.00, 0.40]$. Multiplying by the VAR matrix gives the next horizon $[0.62, 0.26]$, then $[0.39, 0.17]$ - the shock propagates and decays, because the VAR is stable.

The Shock Ripples, Then Fades

A one-unit orthogonalized shock to variable 1 hits it at 1.0 and the sector at 0.4, then both responses decay smoothly toward zero over the following months. The decay is guaranteed by the VAR's stability - shocks die out rather than compounding

Case A: a unit shock to var 1 propagates, then decays to zero (stable VAR)



Why Orthogonalize First

Clean shocks, honest stories

Without orthogonalization

- A shock to one variable is ill-defined.
- You'd double-count the shared movement.
- The response would be uninterpretable.

With the Cholesky

- Each shock is independent and unit-sized.
- The response traces a clean causal-style story.
- Horizons show how long the effect lasts.

Budget the Variance

Where does the uncertainty come from?
Beyond the shape of a response, we want to know how much of a variable's forecast uncertainty each shock explains. The FEVD splits the forecast-error variance into shares that sum to one, horizon by horizon. We compute variable 1's shares - and then see how reordering the variables changes both the Cholesky and the

The question

- How much of var 1's forecast error is its own?
- How much comes from var 2's shock?
- And how does that shift with the horizon?

SW

What we'll do by hand

- Sum the squared responses per shock.
- Divide by the total for the shares.
- Then swap the ordering.

The FEVD

A share of forecast-error variance

$$\text{FEVD}_{ij}(h) = \frac{\text{variance of } i \text{ from shock } j \text{ up to } h}{\text{total forecast-error variance of } i} \quad (\text{shares sum to } 1)$$

How it's built

- Square the responses to each shock.
- Sum them up to the horizon.
- Divide by the grand total.

How to read it

- Each share is a slice of the variance.
- They add to one hundred percent.
- A variance budget, horizon by horizon.

The Variance Shares

Var 1's budget, horizon by horizon

horizon	own shock (var 1)	other shock (var 2)
$h = 1$	100%	0%
$h = 3$	92%	8%
$h = 6$	90%	10%

At one step, all of variable 1's forecast error is its own shock. As the horizon grows, the sector's shock seeps in - reaching about 10% by six months - because the VAR lets the two series interact over time.

The Ordering Problem

A choice that changes the answer

order [1, 2] : var 1 absorbs the shared shock $\neq$ order [2, 1] : var 2 does (answer changes)

Why it happens

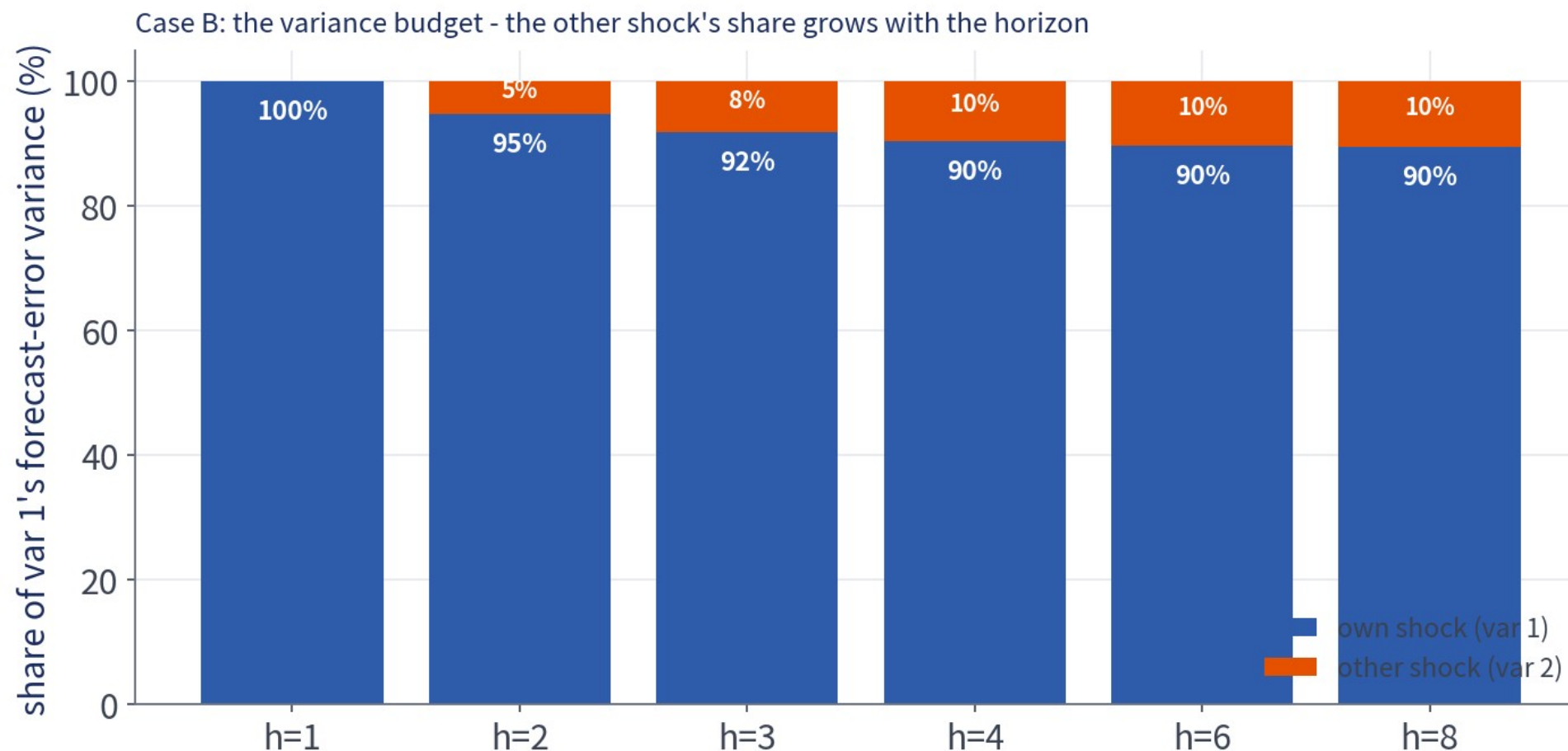
- The Cholesky is triangular.
- The first-ordered variable takes the shared shock.
- So its impact is always the full unit.

What to do

- Order by economic priority - most exogenous first.
- Report how sensitive results are to the order.
- Or use ordering-free alternatives.

The Variance Budget Over Time

At the one-step horizon, variable 1's forecast error is entirely its own shock. As the horizon lengthens, the orange slice - the sector's shock - grows to about a tenth. The stacked bars always sum to 100%: every unit of uncertainty is attributed to some shock



A Budget, With a Warning

Useful, but order-dependent

What the FEVD gives you

- Which shocks drive a variable's uncertainty.
- How that mix shifts across horizons.
- A clear attribution of forecast risk.

The warning

- Both IRF and FEVD depend on the ordering.
- Change the order, change the numbers.
- State your ordering and defend it.

Three Lenses on a VAR

Granger Causality and VAR/VECM together

The week's three tools answer three questions about the same system. Granger causality asks whether one series helps predict another. The impulse response asks how a shock propagates and how long it lasts. The variance decomposition asks how much of each variable's uncertainty each shock explains. Orthogonalization

The response (Case A)

- Cholesky factor $[[1,0],[0.4,0.917]]$.
- Impact $[1.0, 0.4]$, decaying after.
- A clean, orthogonalized shock.

The budget (Case B)

- Own shock 100% at $h=1$, 90% at $h=6$.
- Sector's share grows with the horizon.
- But the ordering sets it all.

Reproduce the Response and the Budget

Direct Claude Code to check your arithmetic

GOAL Orthogonalize, trace the IRF, and compute the FEVD - then swap the ordering.

[INPUT] Prompt · Claude Code

```
Write and run part3_practice.py that, with
A=[[0.5,0.3],[0.1,0.4]] and Sigma=[[1,0.4],[0.4,1]]:
CASE A - IRF:
    * print the Cholesky factor P;
    * trace the IRF to a var-1 shock for h=0..3.
CASE B - FEVD and ordering:
    * print var 1's FEVD shares at h=1,3,6;
    * recompute P for the reversed ordering and show
      the impact response changes.
Then plot the IRF and the stacked FEVD.
```

Check against the slides

- $P = \begin{bmatrix} 1 & 0 \\ 0.4 & 0.917 \end{bmatrix}$.
- IRF $[1.0, 0.4] \rightarrow [0.62, 0.26]$.
- FEVD 100% $\rightarrow$ 90% own.
- reorder flips who absorbs.

Deliverable: part3_practice.py whose printed numbers match the IRF, the FEVD, and the ordering effect.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 Why must you orthogonalize before an IRF?

**Correlated shocks
can't be moved one
at a time**

Q2 What does the FEVD tell you?

**Each shock's share of
a variable's forecast
variance**

Q3 Why does the variable ordering matter?

**The Cholesky gives
the shared shock to
the first variable**

All three come from the Cholesky, the impulse response, and the variance decomposition.

Three Pitfalls to Avoid

Where IRF and FEVD trip people

Common mistakes

- "Shock one variable and hold the others fixed."
- "The IRF is the same for any ordering."
- "A FEVD share is a causal contribution."

The reality

- Correlated shocks must be orthogonalized first.
- Ordering changes both the IRF and the FEVD.
- It's a forecast-variance share, not proven cause.

Two Calculations to Remember

Part 3 applications in one screen

1

Orthogonalize, then trace the response

The Cholesky factor of the correlated covariance was $[[1.0, 0], [0.4, 0.917]]$, giving an impact response of $[1.0, 0.4]$. Pushed through the VAR, it decayed to $[0.62, 0.26]$ then $[0.39, 0.17]$ - a clean shock rippling out and fading.

2

Budget the variance, mind the order

Variable 1's forecast-error variance was 100% its own shock at one step, falling to about 90% by six months as the sector's shock seeped in. But reordering the variables changes the Cholesky - and both the IRF and the FEVD with it.

3

Three lenses on one system

Granger causality tests predictability, the IRF traces a shock's path, and the FEVD budgets the uncertainty. Together they turn a fitted VAR into a full picture of how the variables move each other.

Week 6 complete — from the VAR model, through Granger causality, to impulse responses and variance decomposition.